Quasiperiodic Wang Tiling zk-STARK Signatures (QWTSS): A Post-Quantum Signature Scheme Anchored in the Topologically Frustrated Glassy Phase

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Abstract. The emergence of quantum computing threatens classical public-key cryptography, necessitating the development of robust post-quantum alternatives built upon structurally diverse hardness assumptions. We introduce Quasiperiodic Wang Tiling zk-STARK Signatures (QWTSS), a novel digital signature scheme grounded in the NP-hard combinatorial optimization of bounded aperiodic Wang tile sets. Through stochastic sampling of secret keys within the thermodynamic glassy phase of the configuration space, QWTSS inherently avoids the planted distribution vulnerabilities common to constructive combinatorial primitives. We empirically demonstrate that this critically constrained regime undergoes solution-space shattering and exhibits the Overlap Gap Property (OGP), fracturing into isolated, multifractal solution clusters. The resulting critically constrained operational regime provides extensive combinatorial backtracking against exact classical solvers, while the OGP inhibits heuristic local-search traversal. Furthermore, we model how the phase-space anisotropy of these clusters significantly suppresses the quantum overlap integral. By coupling this structural hardness with a zk-STARK, QWTSS enables a prover to demonstrate knowledge of a specific geometric configuration without leaking the underlying witness. This yields a non-algebraic post-quantum primitive computationally bound to simulated thermodynamic work.

Keywords: Post-Quantum Cryptography · Statistical Mechanics · Wang Tiles · Combinatorial Cryptography · Aperiodic Tilings · zk-STARK

1 Introduction

Post-quantum cryptography relies heavily on lattice-based assumptions like the Shortest Vector Problem (SVP) and Learning With Errors (LWE) [35]. However, practical LWE implementations standardized for global use (such as ML-KEM and ML-DSA) typically do not utilize standard random lattices due to large key sizes. Instead, they rely on structured algebraic variants, such as Module-LWE over polynomial rings [32]. This injection of algebraic structure introduces potential cryptanalytic risk, as future breakthroughs in quantum algorithms could potentially exploit these specific ring geometries [7].

To achieve broader post-quantum resilience, the cryptographic ecosystem requires structural diversity—specifically, primitives that source hardness from non-algebraic, computationally complex domains. QWTSS offers an alternative via bounded Wang tilings [29] over the proven aperiodic Jeandel-Rao set of 11 tiles [17]. This strict aperiodicity precludes the emergence of global periodic symmetries exploitable by the Quantum Fourier Transform. Furthermore, by constraining the system to the thermodynamic glassy phase of the state space, QWTSS establishes strong structural conditions for average-case intractability, providing cryptographic resistance against heuristic algorithms from bounded-error quantum polynomial-time (BQP) adversaries.

To securely externalize this NP-hard problem space for cryptographic application, QWTSS employs a Zero-Knowledge Scalable Transparent Argument of Knowledge (zk-STARK) [5]. The STARK arithmetization naturally accommodates the topological constraints of the grid: the 2D tiling arrangement is unrolled into a 1D execution trace, where local tile-matching rules are encoded as low-degree polynomial transition constraints in the Algebraic Intermediate Representation (AIR). The public key determines the unique boundary constraints of the problem. Because zk-STARKs rely exclusively on symmetric-key primitives (e.g., collision-resistant hash functions), this architecture preserves the post-quantum security of the underlying Wang tile problem without introducing new number-theoretic vulnerabilities, while ensuring sublinear verification time and perfect zero-knowledge of the private grid.

1.1 Our Contributions

In this work, we introduce QWTSS, a novel post-quantum signature scheme grounded in a non-algebraic hardness assumption. To the best of our knowledge, this is the first cryptographic primitive to leverage the topological frustration of bounded aperiodic quasicrystals. Our specific contributions are as follows:

- **A Novel Combinatorial Signature Scheme:** We design QWTSS, a digital signature grounded in the NP-hard bounded Wang tiling problem. By arithmetizing the topological constraints of the Jeandel-Rao 11-tile set into an efficient AIR, we utilize zk-STARKs to achieve non-interactive, sublinear verification of a 2D quasicrystalline lattice without leaking the private configuration.
- **Empirical Validation of the Overlap Gap Property (OGP):** We formulate key generation as a constrained thermodynamic optimization problem, demonstrating that the operational glassy phase is critically constrained and exhibits solution-space shattering. By providing compelling evidence for the OGP, we establish strong structural conditions for average-case intractability and robust cryptographic resistance to multiple categories of classical solvers.
- **Quantum Overlap Suppression via Phase-Space Anisotropy:** We model how the combinatorial anisotropy of QWTSS solution clusters—which form degenerate, fractal-like sub-spaces—significantly suppresses the quantum overlap integral, degrading global quantum tunneling and QAOA fidelity.
- **Comprehensive State Space Characterization:** Using Annealed Importance Sampling (AIS), modified Rosenbluth-Knuth estimators, and greedy trajectory solvers, we provide a rigorous volumetric and geometric mapping of the underlying state space.
- **Empirical Bounds on Structural Leakage:** We formalize the Labbé Oracle Attack to evaluate the scheme's vulnerability to deterministic boundary-projection cryptanalysis, demonstrating that QWTSS retains a significant post-quantum security margin even under a strong threat model equipped with certain artificial advantages.
- **High-Performance Reference Implementation:** We introduce an open-source C++ reference implementation featuring a custom MCMC solver with optional GPU acceleration and seamless integration with the StarkWare Stone Prover[1].

[1] Reference implementation and supporting code are available at https://github.com/axonetric/qwtss

2 Background

Before detailing the specific architecture of the QWTSS scheme, it is necessary to establish the theoretical foundations upon which its security and functionality rely. This section reviews the mathematical complexity of bounded Wang tilings, the thermodynamic principles underlying the state space, the Overlap Gap Property, the zk-STARK innovations, and previous research relevant to this paper.

2.1 Wang Tiling and NP-Hardness

The study of aperiodic tilings began with Hao Wang's seminal work on the Domino Problem in 1961 [40]. Wang tiles are unit squares with colored edges. A valid tiling of a 2D plane requires that all adjacent tiles share the same edge color, and tiles may not be rotated.

The theoretical security of our system relies on the Bounded Domino Problem: Given a finite set of Wang tiles $\mathscr{A}$ and a specific boundary constraint, can an $L \times L$ grid be validly tiled? In 1966, Berger proved that the general Domino Problem (tiling the infinite plane) is undecidable by demonstrating that a set of Wang tiles can simulate the execution tape and state transitions of any universal Turing machine [6]. Because finding a valid infinite tiling is equivalent to the halting problem, it naturally follows that the computationally bounded variant—determining if a finite $L \times L$ grid can be tiled subject to fixed boundary conditions—simulates a bounded non-deterministic Turing machine, placing the decision problem firmly in the NP-complete complexity class. Consequently, the task of actually discovering a valid configuration constitutes an NP-hard search problem.

For cryptographic utility, we require the alphabet $\mathscr{A}$ to be aperiodic—meaning it can tile the plane, but only in non-repeating, complex configurations. As demonstrated by Kari [18], the propagation of aperiodic tiles behaves analogously to dynamic systems and chaotic flows. Furthermore, Levitov established that in such non-periodic systems, local matching rules strictly dictate the emergence of complex, unique global structures [28]. Following Robert Berger's discovery of the first aperiodic set of Wang tiles in 1966—which famously required an alphabet of 20,426 tiles—progressively smaller aperiodic sets were discovered over the subsequent decades [37, 3, 18, 10]. This reduction culminated in 2015, when Jeandel and Rao discovered the proven minimal 11-tile set via an exhaustive computer search [17].

However, the definitive topological characterization of this specific alphabet was formalized later by Sébastien Labbé. Labbé proved that the zero-defect configurations of the Jeandel-Rao 11-tile set (JR-11) possess a strict substitutive structure. Specifically, he demonstrated that the tilings are generated by a hierarchical substitution rule where specific finite "patches" of tiles recursively map to larger, self-similar macro-tiles [23]. Furthermore, he showed that these tilings correspond exactly to the coding of a $\mathbb{Z}^2$-action by rotations on a two-dimensional torus, successfully constructing a Markov partition for the JR-11 tile set [22].

The existence of these exact tiling solutions necessitates a departure from the perfect tiling case for cryptographic security. We observe that the quasiperiodic nature of zero-defect tilings enforces topological rigidity, locking the grid's interior structure to its boundary state. In this regime, the conditional joint entropy collapses toward zero; the number of valid interior solutions associated with any given perimeter (public key) is structurally bounded and does not scale with the area of the grid. Consequently, we require that candidate tiling solutions (private keys) in the QWTSS scheme contain a strictly positive number of geometric defects (mismatched colored edges), as the inclusion of entropy-carrying defects rapidly restores a super-polynomial search space even under fixed boundary conditions.

2.2 Thermodynamic Formulation

While the traditional Bounded Domino Problem is framed as a binary decision problem (valid or invalid), we reformulate it as an optimization problem under the laws of statistical mechanics.

We treat the $L \times L$ grid as a thermodynamic system. The configuration space is governed by a Hamiltonian energy function, E, defined simply as the total number of geometric defects in the grid. Robinson established the thermodynamic limit of Wang tilings, demonstrating a formal mathematical relationship between the "energy" (defects) of a tiling configuration and its global structural stability [36].

As a Markov Chain Monte Carlo (MCMC) solver navigates this space, lowering the annealing temperature forces the grid to crystallize, triggering a glassy phase transition at a critical threshold. The system experiences a kinematic slowdown, where a progressively increasing expenditure of computational cycles is required to reach progressively lower ground states. Because the selected Wang tile set is strictly aperiodic, attempting to greedily satisfy local constraints forces non-local structural contradictions. Resolving a single local defect frequently requires cascading rearrangements of non-local tiles. Recent advancements in the statistical mechanics of Wang tilings by Canfora and Cedeno support this, demonstrating that Wang tiling alphabets possessing topological frustration exhibit transitions to chaotic regimes that dynamically arrest polynomial-time solutions [8].

By enforcing specific geometric constraints, we successfully preclude a narrow subset of synthetic deterministic solutions. This ensures not only that valid private keys ($\mathcal{SK}$) are securely sampled from the target QWTSS stationary distribution, but also that brute-force forgery attempts necessitate a substantial expenditure of simulated thermodynamic work. Because tilings with low defect counts behave as a highly rigid, long-range ordered quasicrystal, the underlying dynamical system ensures that an attacker cannot exploit purely localized heuristics or bypass the structural dependencies imposed by the public key ($\mathcal{PK}$) boundary upon the grid's interior. Consequently, we leverage this topological frustration as a defense multiplier to enhance the scheme's cryptographic preimage resistance.

2.3 Statistical Physics of Constraint Satisfaction

The security of random aperiodic tiling grids is mathematically reinforced by the statistical mechanics of Constraint Satisfaction Problems (CSPs). It is well-documented that random NP-hard problems exhibit a sharp computational phase transition between under-constrained (highly satisfiable) and over-constrained (unsatisfiable) regimes [19]. Algorithmic hardness does not peak at these extremes; rather, it diverges precisely at the critical boundary where the density of valid solutions becomes vanishingly sparse but remains strictly positive.

At this critical threshold, the global effective branching factor of the search tree approaches $b_{\text{eff}} = e^{\epsilon}$, where ϵ represents a sufficiently small positive residual topological entropy per variable. As the system enters this critically constrained region, the contiguous manifold of valid solutions undergoes a topological transformation known as *solution space shattering* [1, 31]. In this shattered regime, the state space fractures into an exponentially large number of isolated, disjoint clusters separated by extensive Hamming distances [21].

This geometric shattering is formalized computationally by the Overlap Gap Property (OGP) [13]. OGP establishes a rigorous topological barrier within the solution space of near-critical optimization problems. It demonstrates that for random instances of CSPs, globally valid and near-optimal solutions do not form a continuous, traversable gradient. Under OGP, any two near-optimal configurations either (1) share a large proportion of their structure (residing within the same cluster), or (2) differ radically (lying in entirely different clusters), separated by intermediate "overlap" states that are statistically forbidden or vanishingly rare.

This topological isolation strongly resists local-search heuristics [14]. When a local solver encounters a structural conflict, it cannot simply execute a small, localized adjustment to bypass the violation. Because the intermediate states bridging two valid clusters are highly defective, the algorithm is trapped in stable local minima, forced to backtrack and dismantle large portions of its already developed partial solution. As established in recent literature, OGP formalizes an intractable barrier for robust classes of advanced solvers, including MCMC, approximate message passing (AMP), and low-degree polynomials [41, 13].

Concurrently, the underlying critically constrained regime inhibits exact deterministic solvers. For Conflict-Driven Clause Learning (CDCL) algorithms attempting to navigate problem instances within this regime, the resolution tree size is proven to scale exponentially as $2^{\Omega(N)}$. The solver is forced to search prohibitively deep into the decision tree before discovering a logical contradiction, preventing early branch pruning and maximizing backtracking depth.

Crucially, this framework provides a compelling theoretical foundation for structural average-case intractability in our cryptographic scheme. While traditional NP-hardness only guarantees that the *worst-case* instances of a problem are difficult, OGP and solution space shattering are widely recognized as the primary mathematical origins of algorithmic hardness in *random, average-case* CSPs. With explicit calibration of the QWTSS operational parameters, we provide strong empirical evidence in Section 5 demonstrating both the OGP and phase criticality. This implies that the scheme retains average-case computational intractability against a wide range of powerful algorithmic families.

2.4 Zero-Knowledge STARKs

STARKs provide a post-quantum alternative to SNARKs, eliminating trusted setups by relying on symmetric primitives like collision-resistant hash functions [5]. Sublinear verification is achieved via the Fast Reed-Solomon Interactive Oracle Proof of Proximity (FRI) [4]. To utilize this proof system, the combinatorial logic of the Wang tile grid must be arithmetized into an AIR. The AIR flattens the 2D grid topology into a 1D execution trace, governed by two distinct types of low-degree polynomial equations. *Transition constraints* enforce the local edge-matching rules while tracking the spatial distribution and cryptographic fingerprint of the grid defects across the trace. Conversely, *boundary constraints* map the user's public key to the grid's perimeter and enforce certain final thresholds.

While traditional combinatorial zero-knowledge proofs rely on extensive Merkle tree authentication paths that scale linearly with the number of revealed views, the FRI protocol algebraically bounds the proof validation to a logarithmic payload. By mapping the 64×64 grid size to a compact trace length, the QWTSS arithmetization enables highly memory-efficient key generation and low-latency verification on standard consumer hardware. For the proof-of-concept implementation of QWTSS, we leverage the open-source Stone Prover released by StarkWare [39], which provides a highly optimized, production-grade C++ proving engine.

2.5 Related Work

Wang Tilings and Cryptography. The theoretical intersection of Wang tilings and cryptography dates to the early 2000s. In 2003, Levin established the bounded Wang tiling problem as a combinatorial complete one-way function, providing the foundation for tiling-based primitives [27]. Concurrently, the formalization of the abstract Tile Assembly Model (TAM) demonstrated that the edge-matching rules of Wang tiles could be physically emulated by DNA self-assembly [42]. While early TAM research explored using this physical, molecular self-assembly as a highly parallel mechanism to attack existing RSA and Diffie-Hellman cryptosystems, our architecture inverts this paradigm. We utilize the thermodynamic properties and dynamic arrest of aperiodic Wang tile crystallization not as an attack vector, but as a fundamental security barrier of the cryptosystem itself.

Zero-Knowledge NP-Hard Signatures. Conceptually, the QWTSS architecture functions as a post-quantum, hash-based signature scheme derived from an NP-hard combinatorial optimization problem. The classic Goldreich-Micali-Wigderson (GMW) protocol for Graph 3-Coloring established the blueprint for proving NP-complete solutions without leakage [15]. The design intuition of our scheme shares a direct theoretical lineage with GMW. In the GMW protocol, a prover masks a graph, and a verifier probabilistically challenges adjacent vertices to ensure they possess different colors. Similarly, our scheme proves the validity of adjacent Wang tile edge alignments. However, rather than relying on interactive commit-and-reveal queries, QWTSS modernizes this foundational paradigm. We embed the geometric and topological constraints of the 2D grid into an AIR, utilizing non-interactive zk-STARKs to achieve sublinear verification of the combinatorial proof and a significantly more compact signature.

The recently proposed Eidolon scheme [9] similarly constructs signatures by applying the Fiat-Shamir transform to a generalized GMW protocol for k-colorability. However, while Eidolon relies on Merkle-tree vector commitments to selectively reveal combinatorial views, QWTSS translates the geometric problem entirely into the algebraic domain by arithmetizing the problem space and underlying rules into an AIR and leveraging zk-STARKs.

3 QWTSS Specification

This section formalizes the architecture of the QWTSS scheme. We define the deterministic generation of the public boundary, the thermodynamic derivation of the private grid, and the AIR constraints that form the zero-knowledge signature.

3.1 The Aperiodic Tile Set (JR-11)

Before defining the key generation mechanics, we must formalize the combinatorial alphabet underpinning the scheme. QWTSS employs the Jeandel-Rao (JR-11) Wang tile set, an aperiodic set comprising 11 distinct tiles. While the literature frequently describes JR-11 as a 4-color set based on its theoretical minimum, the original computational output discovered by Jeandel and Rao utilized five distinct edge colors (typically labeled 0 through 4) [17]. Although Jeandel and Rao demonstrated that colors 0 and 4 can be merged without compromising aperiodicity, QWTSS intentionally implements the unmerged, 5-color palette, aligning with conventions established in subsequent research.

All empirical data and theoretical bounds presented in this paper are derived from this 5-color JR-11 implementation. We also briefly evaluated one other small aperiodic alphabet—the Ammann 16-tile, 6-color set (A-16) [3]—as an alternative candidate; however, JR-11 was selected because it exhibits higher joint entropy under identical boundary conditions, as well as superior topological defect resilience. While other small aperiodic Wang tile sets (comprising fewer than 100 tiles) exist, they remain less exhaustively analyzed in the broader literature, making them less suitable for establishing foundational cryptographic hardness.

Furthermore, because the zero-defect state of JR-11 has been rigorously formalized by Labbé, selecting this set allows us to preemptively model the scheme's susceptibility to deterministic oracle attacks (detailed in Section 7). Given that Labbé recently achieved a similar mathematical formalization for the zero-defect state of the A-16 set [24, 25], it is prudent to design under the assumption that the zero-defect configurations of other aperiodic tile sets may eventually be deterministically mapped. By anchoring our security model to a set whose deterministic properties are already fully understood, we ensure the scheme's baseline resilience against future cryptanalytic breakthroughs.

3.2 The Public Key and NUMS Boundary Generation

Unlike traditional cryptosystems where the public key is a discrete algebraic object, a QWTSS public identity is a cryptographic tuple that dictates a commitment to a valid, boundary-constrained two-dimensional $L \times L$ Wang tiling grid. We define the standard grid dimension as $L = 64$. Unless otherwise specified, all subsequent data and analyses in this work are conducted within this 64×64 lattice.

A valid QWTSS public identity record is defined as the tuple:

$$\mathcal{PK} = \{\text{Username}, \text{Identity\_Nonce}, \text{Grid\_Fingerprint}, VB\}$$

where Username is the plaintext identity of the user, Identity_Nonce is a public salt used for key rotation or revocation, Grid_Fingerprint is a cryptographic commitment to the grid's internal tiling arrangement, and VB is a version byte for versioning and interoperability. Conceptually, $\mathcal{PK}$ functions as a macroscopic public commitment to a unique, identity-bound quasicrystalline lattice.

Spliced NUMS Boundary Derivation. To ensure the public key is a Nothing-Up-My-Sleeve (NUMS) value and prevent certain types of attacks, the 64×64 grid perimeter (comprising up to 256 pinned edge colors) is generated strictly deterministically from the user's identity. However, generating a purely random boundary is thermodynamically pathological; it either prevents the MCMC annealer from converging to the required target defect band, or forces highly skewed defect distributions near the border. Conversely, a purely non-random boundary, such as one obtained from an unconstrained annealing process, overly reduces the complexity of the search space. To inject a reasonable amount of topological frustration required for cryptographic hardness, we employ a deterministic oracle splicing technique. The generation proceeds as follows:

1. A 256-bit cryptographic seed is derived via $S = \mathrm{SHA256}(\text{Username} \parallel \text{`|'} \parallel \text{Identity\_Nonce} \parallel \text{`|'} \parallel VB)$.
2. This seed S initializes a deterministic Cryptographically Secure Pseudo-Random Number Generator (CSPRNG), specifically instantiated using the ChaCha20 stream cipher.
3. The CSPRNG drives the $O(1)$ Labbé oracle consultation [22, 26] to generate two independent, zero-defect 64×64 reference grids, G_1 and G_2.
4. The CSPRNG generates a 1D perimeter walk mask containing exactly 14 splice points. This mask deterministically splices the boundary of G_1 and G_2 in a strictly alternating sequence, resulting in a composite boundary that inherits $\approx 50\%$ of its edge tiles from G_1 and $\approx 50\%$ from G_2. The 14 spliced segments are generated with a mathematically guaranteed minimum length to prevent trivial segment lengths, while the CSPRNG supplies stochastic jitter to distribute the remaining lengths and assigns a random index for the start of the perimeter walk.
5. Finally, the CSPRNG applies a deterministic dropout mask based on the *boundary pinning density* ($\rho_B = 0.70$). This parameter dictates exactly which of the 256 outward-facing perimeter edge colors are strictly pinned as public constraints, and which are left unconstrained. The resulting array of constrained edges is defined as the Pinned_Boundary_Colors.

This spliced boundary guarantees the absence of a trivial zero-defect interior, forcing the MCMC annealer to perform stochastic computational work to resolve the resulting topological conflicts, while also avoiding the failed or degenerate results associated with completely randomized borders.

Boundary Pinning Density (ρ_B). As introduced during the public key generation protocol, $\rho_B \in (0, 1]$ denotes the boundary pinning density—the fraction of grid perimeter colors strictly retained as a public key constraint. A $\rho_B = 1.0$ dictates a fully 100% constrained perimeter. However, for certain empirical reasons outlined in Section 4, it is advantageous to introduce a partial perimeter relaxation. In the QWTSS specification, we assign $\rho_B = 0.70$.

3.3 The Private Key and Grid Fingerprint

While the public key defines the boundary of the system, the private key $\mathcal{SK}$ consists of the full 64×64 grid of JR-11 Wang tiles $\mathcal{A}$ that satisfies all $\mathcal{PK}$-derived constraints within the specified energy (defect) range.

MCMC Annealing and State Discovery. The private key is discovered through a simulated annealing process utilizing an MCMC sampler. Given the spliced NUMS boundary, the sampler explores the boundary-conditioned configuration space to find a state where the total number of geometric defects D falls within the scheme's target band (representing $\approx$ 1.98% to 2.11% of the 8,064 interior edges):

$$160 \leq D \leq 170$$

Because the configuration space possesses high joint entropy ($\approx$ 940 bits) despite the grid being roughly 98% perfectly crystallized, the specific arrangement of these defects is effectively unique to each annealing run (Figure 1). Crucially, the underlying spatial MCMC state transitions must preserve instantaneous detailed balance at each temperature step. Preserving this spatial detailed balance ensures that the solver's exploration remains unbiased, allowing the final sampled private keys to closely approximate the uniform stationary distribution of the target defect band. This precludes planted distribution vulnerabilities or algorithmic sampling biases that could artificially reduce the system's joint entropy.

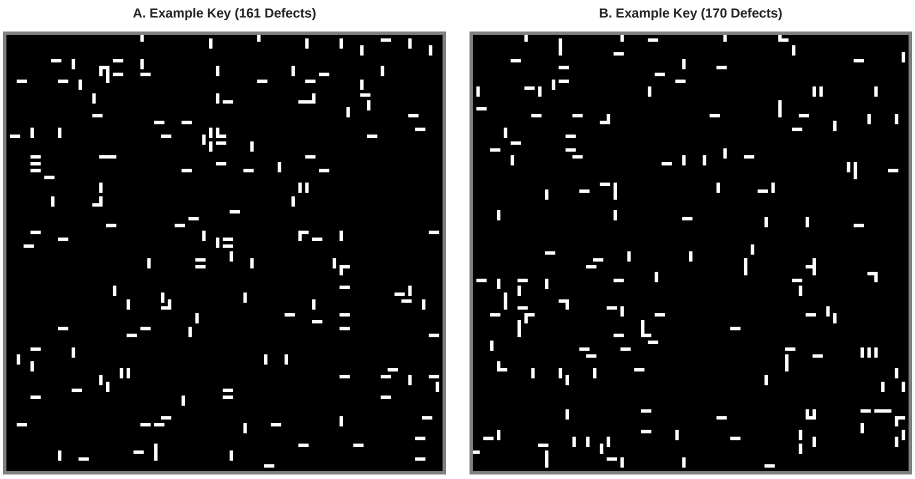

Figure 1: Two examples of the edge mismatch (defect) distributions produced by the $\mathcal{SK}$ annealing process, where $D \in [160, 170]$ (edges enlarged for visibility).

The Grid Fingerprint. To bind a unique private key to the public identity without revealing the tile configuration, we derive a Grid_Fingerprint. This is a succinct cryptographic commitment to the grid's internal JR-11 tile configuration. The fingerprint is computed via a continuous, dense cryptographic accumulator integrated directly into the STARK execution trace. The Poseidon hash is specifically chosen for its algebraic efficiency within STARK execution traces, requiring a low polynomial degree to express its S-box transitions. We elaborate on the security of the Poseidon hash in Appendix A.

1. The 64×64 grid is flattened into a 1D execution trace of 4,096 sequential steps (raster-scan order).

2. At each step i, the Poseidon hash state absorbs the specific JR-11 tile identifier (tile_id_i) placed at that coordinate, along with the current trace step index (i) to enforce cryptographic domain separation.

3. The state continuously updates via the algebraic transition:

$$\text{hash}_{i+1} = \text{Poseidon}(\text{hash}_i, \text{tile\_id}_i, i) \tag{1}$$

4. After processing all 4,096 tiles, a 504-bit digest is squeezed from the rate elements of the terminal hash state to form the Grid_Fingerprint.

The Pigeonhole Principle. Within the 64×64 Wang tile grid, there are ≈ 940 bits of boundary-conditioned joint entropy (given an arbitrary $\mathcal{PK}$). Because the scheme compresses this large geometric state space into a single 504-bit Poseidon digest, it is fundamentally governed by the Pigeonhole Principle: an exponentially large number of valid private key grids ($\mathcal{SK}$) must collide on any given hash. However, the security of the public identity strictly relies on the preimage resistance of the Poseidon permutation. We must carefully distinguish between the effort required to satisfy the public boundary constraints alone versus forging a complete $\mathcal{PK}$ identity. While generating *any* topologically valid grid that satisfies the perimeter constraints of the {Username, Identity_Nonce} tuple is easily accomplished (requiring only ≈ 0.5 seconds of GPU-accelerated annealing), discovering a topologically valid grid that simultaneously satisfies the perimeter constraints *and* produces an identical 504-bit grid fingerprint is computationally intractable. Therefore, the grid fingerprint serves as an immutable commitment that the prover holds the exact, topologically valid interior they originally sampled, anchoring the scheme's forgery resistance.

Private Key Generation (KeyGen). QWTSS is presented as a concrete cryptographic instantiation targeting a fixed post-quantum security level achieved through the globally fixed parameters pp: a grid size of $L = 64$, an operational defect band of $D \in [160, 170]$, a boundary pinning density $\rho_B = 0.70$, and a 504-bit grid fingerprint.

Algorithm 1 KeyGen(pp)

Require: Username, Identity_Nonce, VB
Ensure: Public key $\mathcal{PK}$, Secret key $\mathcal{SK}$

1: $S \leftarrow$ SHA256(Username ‖ '|' ‖ Identity_Nonce ‖ '|' ‖ VB)
2: $B \leftarrow$ GenerateSplicedBoundary(S, ρ_B) ▷ Deterministic NUMS boundary
3: $G \leftarrow$ MCMC_Anneal(B, Target_$D \in [160, 170]$) ▷ Sample from glassy phase
4: $F \leftarrow$ Poseidon_Trace_Digest(G) ▷ 504-bit trace fingerprint
5: $\mathcal{PK} \leftarrow$ {Username, Identity_Nonce, F, VB}
6: $\mathcal{SK} \leftarrow G$
7: **return** ($\mathcal{PK}, \mathcal{SK}$)

3.4 The Algebraic Intermediate Representation (AIR)

To construct the zero-knowledge STARK proof, the 2D constraint satisfaction problem of the QWTSS grid must be arithmetized into a system of polynomial equations over a finite prime field $\mathbb{F}_p$. We achieve this by mapping the $L \times L$ Wang tile grid into the first $N = L^2$ steps of a larger 1D execution trace, defining localized transition polynomials over this valid domain while reserving the remainder of the trace for zero-knowledge blinding noise. The purpose of the AIR is to fully encapsulate and formalize every constraint within the QWTSS scheme in a non-interactive fashion. This ensures that an adversary is mathematically incapable of forging a signature or bypassing the required computational work—even if utilizing a custom prover implementation—all while retaining perfect zero-knowledge of the private data.

One-Hot Trace Mapping and Color Derivation. To avoid the high-degree domain polynomials required to encode tile IDs as single field elements, our AIR employs a one-hot boolean encoding that minimizes polynomial degree. Let $\mathcal{A}$ be the aperiodic alphabet. For each step k in the execution trace, the state includes $|\mathcal{A}|$ columns, denoted $b_{i,k}$ for $i \in [0, |\mathcal{A}| - 1]$. Let s_{exe} be a periodic selector that is 1 inside the valid grid ($k < L^2$) and 0 in the subsequent trace padding.

The trace enforces exactly one active tile per spatial cell via boolean and exclusivity constraints bounded by this selector:

$$s_{exe} \cdot b_{i,k} \cdot (b_{i,k} - 1) = 0 \quad \forall i \tag{2}$$

$$s_{exe} \cdot \left(\sum_{i=0}^{|\mathcal{A}|-1} b_{i,k} - 1 \right) = 0 \tag{3}$$

Using this boolean representation, the physical edge colors of the active tile at step k (denoted $d_{top,k}$, $d_{right,k}$, $d_{bottom,k}$, $d_{left,k}$) are derived dynamically via linear combinations:

$$d_{top,k} = \sum_{i=0}^{|\mathcal{A}|-1} b_{i,k} \cdot \mathcal{A}[i]_{top} \tag{4}$$

The remaining three edge colors are derived identically using their respective physical constants.

Transition Constraints (Edge Matching and Defects). We introduce boolean columns for horizontal and vertical edge defects, h_k and v_k. The trace padding region ($k \geq L^2$) is intentionally unconstrained by s_{exe} and is populated with cryptographically secure random blinding noise, a requirement for the FRI protocol to achieve perfect zero-knowledge without leaking the private grid configuration. The selectors s_h and s_v selectively deactivate constraints at the right and bottom grid boundaries, respectively. The edge-matching polynomials are defined as:

$$s_h \cdot (d_{right,k} - d_{left,k+1}) \cdot (1 - h_k) = 0 \tag{5}$$

$$s_v \cdot (d_{bottom,k} - d_{top,k+L}) \cdot (1 - v_k) = 0 \tag{6}$$

The defect values are constrained to strictly boolean states $\{0, 1\}$. To mathematically enforce the scheme's target defect band ($160 \leq D \leq 170$) without requiring additional trace columns, we introduce a boolean slack variable, $dummy_k$. We restrict the maximum allowable slack to exactly 10 by defining a periodic selector, s_{slack_limit}, which evaluates to 0 for the first 10 steps of the trace and 1 for the remainder of the valid grid execution. The slack limit polynomial is defined as:

$$s_{slack\_limit} \cdot dummy_k = 0 \tag{7}$$

A running accumulator column a_k sequentially tracks the total combined defect and slack count across the valid trace steps:

$$s_{exe} \cdot (a_{k+1} - (a_k + h_k + v_k + dummy_k)) = 0 \tag{8}$$

Because the slack variable can only contribute a maximum of 10 artificial defects, constraining the accumulator to exactly 170 at the terminal grid step s_{last} strictly guarantees that the true geometric defect count falls within the designated $[160, 170]$ band:

$$s_{last} \cdot (a_N - 170) = 0 \tag{9}$$

Boundary Constraints (Partial Pinning). The public identity dictates the required boundary edge colors of the grid. To support partial boundary pinning—allowing a subset of the perimeter to remain unconstrained—we dynamically mute the boundary constraints within the AIR. Let $v_{north,k}$, $v_{south,k}$, $v_{west,k}$, and $v_{east,k}$ be periodic columns encoding the public key's 1D boundary arrays. We define geometric periodic selectors (s_{north}, s_{south}, s_{west}, s_{east}) that isolate the physical perimeter of the grid. Crucially, if a specific perimeter coordinate is designated as un-pinned by the public key's boundary dropout mask, the corresponding selector evaluates to 0. This mathematically disables the constraint at that exact geometric coordinate without increasing the algebraic degree of the system. The boundary constraints are enforced algebraically against the dynamically derived tile colors:

$$s_{north} \cdot (d_{top,k} - v_{north,k}) = 0 \tag{10}$$

$$s_{south} \cdot (d_{bottom,k} - v_{south,k}) = 0 \tag{11}$$

$$s_{west} \cdot (d_{left,k} - v_{west,k}) = 0 \tag{12}$$

$$s_{east} \cdot (d_{right,k} - v_{east,k}) = 0 \tag{13}$$

Poseidon Hash State (The Grid Fingerprint). Finally, the execution trace continuously absorbs the defect configuration into a Poseidon hash state column H_k, validating the degree-3 non-linear S-box transitions at each step. To secure the trailing edge of the computation against algebraic forgery, the hash state continues for 64 additional *blanking rounds* beyond the spatial grid ($k \in [N, N+63]$). During these rounds, the spatial selector s_{exe} deactivates, and a zero-lock constraint forces the absorbed tile ID strictly to 0. A terminal selector, $s_{identity\_last}$, anchored at the final blanking step, rigorously constrains the hash state to equal the public commitment:

$$s_{identity\_last} \cdot (H_{N+63} - \text{Grid\_Fingerprint}) = 0 \tag{14}$$

By enforcing this system of low-degree polynomials, the resulting Fast Reed-Solomon Interactive Oracle Proof of Proximity (FRI) mathematically proves possession of a specific, valid lattice interior anchored to the public identity.

3.5 Advanced Thermodynamic and Spatial Constraints

While the baseline AIR ensures geometric validity and enforces forgery resistance, an impatient prover could shortcut the intended computational difficulty by clustering defects, creating continuous defect walls, or using known zero-defect solutions to avoid the simulated thermodynamic work required to generate a valid private key ($\mathcal{SK}$). We detail these circumvention strategies—and their corresponding constraints—in Appendix H. To enforce the intended computational effort of key generation, we introduce three additional trace constraints. These constraints serve three distinct purposes: (1) to preclude provers from generating private keys quickly and deterministically, thereby preventing the use of heavily biased and cryptographically insecure $\mathcal{SK}$ distributions; (2) to enforce explicit topological quality control by automatically excluding degenerate, simplistic, or structurally disjoint grid configurations; and (3) to establish a rigorous foundation for Proof-of-Work (PoW) extensions of the QWTSS scheme, which we explore separately in a forthcoming companion paper.

Zero-Knowledge Alien Tile Subsets. The scheme requires the inclusion of a specific minimum quantity of "alien" tiles, which are tile IDs that differ from the tiles occupying the corresponding coordinates in both $\mathcal{PK}$-derived oracle grids G_1 and G_2. This guarantees problem complexity, and a large number of alien tiles naturally emerge during the stochastic annealing process. To enforce this lower bound on the alien tile count without revealing the private tile IDs, the AIR utilizes a zero-knowledge algebraic inverse trick. Let $alien\_acc_k$ track the number of alien tiles placed up to step k. The transition difference, $\Delta alien_k = alien\_acc_{k+1} - alien\_acc_k$, is strictly bound to $\{0, 1\}$.

If the prover claims an alien tile ($\Delta alien_k = 1$), they must provide a valid field inverse, $alien\_inv_k$, demonstrating that the private trace tile ID id_k matches neither $G_{1,k}$ nor $G_{2,k}$. If id_k matches either oracle tile at step k, the product $(id_k - G_{1,k})(id_k - G_{2,k})$ evaluates to 0, rendering the inverse mathematically impossible to calculate and failing to satisfy the polynomial constraint:

$$s_{exe} \cdot \Delta alien_k \cdot \big((id_k - G_{1,k})(id_k - G_{2,k}) \cdot alien\_inv_k - 1\big) = 0 \tag{15}$$

The total alien tile count is then rigidly anchored at the boundary step to ensure the required minimum threshold is met:

$$s_{last} \cdot (alien\_acc_N - \text{Target\_Alien\_Count}) = 0 \tag{16}$$

Core32 Spatial Defect Density. To prevent provers from artificially clustering defects extensively near the grid's border regions—a recurring theme in many circumvention strategies—we enforce a localized density requirement within the central 32×32 core of the grid. We define a spatial periodic selector s_{core32} that evaluates to 1 exclusively within this inner boundary.

Let δ_k be a derived boolean that equals 1 if the tile at step k contains any defective edges, and 0 otherwise. A dedicated accumulator, $core32\_acc_k$, tracks these core defects. The transition is constrained such that the accumulator may only increment if the spatial selector is active and the tile is defective:

$$s_{exe} \cdot \Delta core32_k \cdot (s_{core32} \cdot \delta_k - 1) = 0 \tag{17}$$

A final boundary constraint ensures the core accumulator meets the minimum required density threshold, forcing the computational work into the hardest region of the grid.

Orthogonal Seam Defect Limits (Line_Max). To penalize synthetic, contiguous defect walls (decoupling attacks) that artificially partition the grid, the AIR caps the maximum allowable defects along any single horizontal or vertical seam. Because traditional range checks (e.g., $x \leq \text{Line\_Max}$) require expensive binary decompositions in STARKs, we implement highly efficient orthogonal slack accumulators.

Let $h\_seam\_acc_k$ track the defects across a horizontal row. Instead of directly proving the total is less than or equal to Line_Max, the prover is forced to inject invisible dummy defects into the seam until the accumulator reaches exactly Line_Max. The transition polynomial enforces that the step difference minus the actual physical defect v_k is boolean (either 0 or a 1 for the dummy slack):

$$s_v \cdot s_h \cdot \Delta h_k \cdot (\Delta h_k - 1) = 0 \quad \text{where} \quad \Delta h_k = h\_seam\_acc_{k+1} - h\_seam\_acc_k - v_k \tag{18}$$

To ensure the final tile of the row cannot be exploited as a blind spot, we utilize an inclusive end anchor. At the final column of the row (where $s_h = 0$), the polynomial directly evaluates the remaining gap to the limit:

$$s_v \cdot (1 - s_h) \cdot \Delta_{end} \cdot (\Delta_{end} - 1) = 0 \quad \text{where} \quad \Delta_{end} = \text{Line\_Max} - h\_seam\_acc_k - v_k \tag{19}$$

This architecture strictly bounds the defects per seam on both the X and Y axes simultaneously without adding any new trace polynomials or increasing the algebraic degree of the system.

3.6 Signature Generation and Verification

While the underlying proof system relies on standard zk-STARK mechanics, QWTSS implements a Strong Fiat-Shamir transformation to cryptographically bind the arbitrary message to the public identity and the execution trace. Let $\mathcal{R}$ denote the NP-hard relation defining a valid 64×64 JR-11 Wang Tiling configuration satisfying the AIR constraints detailed in Sections 3.4 and 3.5.

Signature Generation (Sign). Because the initial Fiat-Shamir seed strictly binds the message and all public boundary parameters, altering the message or submitting the proof against a different public key entirely alters the subsequent pseudo-random FRI challenge queries. This renders the proof π invalid and nullifies Weak Fiat-Shamir malleability.

Algorithm 2 Sign($\mathcal{SK}, \mathcal{PK}, m$)

Require: Secret key $\mathcal{SK} = G$, Public key $\mathcal{PK}$, Message $m \in \{0, 1\}^*$
Ensure: Signature σ

$seed_{FS} \leftarrow \text{SHA256}(\mathcal{PK} \parallel m)$ ▷ Fiat-Shamir challenge root
Statement $x \leftarrow \mathcal{PK}$
Witness $w \leftarrow G$ ▷ The private tile configuration
$Trace \leftarrow \text{Generate\_Trace}(w)$ ▷ Includes blinding noise for $k \geq L^2$
$\pi \leftarrow \text{STARK.Prove}(x, Trace, seed_{FS})$ ▷ Prove $(x, w) \in \mathcal{R}$
return $\sigma = \pi$

Signature Verification (Verify). The verifier reconstructs the boundary constraints deterministically from the public identity tuple and validates the STARK proof against the corresponding Algebraic Intermediate Representation.

Algorithm 3 Verify($\mathcal{PK}, m, \sigma$)

Require: Public key $\mathcal{PK}$, Message m, Signature $\sigma = \pi$
Ensure: Accept (1) or Reject (0)

$seed_{FS} \leftarrow \text{SHA256}(\mathcal{PK} \parallel m)$ ▷ Reconstruct challenge root
$B \leftarrow \text{GenerateSplicedBoundary}(\mathcal{PK})$ ▷ Reconstruct NUMS boundary
Statement $x \leftarrow \mathcal{PK}$
$valid \leftarrow \text{STARK.Verify}(x, \pi, seed_{FS})$
if $valid$ = TRUE **then**
  return 1
else
  return 0
end if

4 Empirical Analysis

4.1 Boundary-Conditioned Joint Entropy

Initial attempts to quantify the conditional joint entropy of the state space in the glassy phase using standard constant-temperature MCMC methods proved intractable. The local rigidity and long-range order imposed by the aperiodic tilings create energy barriers that prevent ergodicity. To overcome these challenges and directly evaluate the partition function, we employ Annealed Importance Sampling (AIS) [33], which operates on the principles of non-equilibrium thermodynamics. AIS provides a mathematically rigorous stochastic Monte Carlo estimator for the ratio of partition functions between a tractable, unconstrained base distribution and the highly constrained target distribution (Appendix B). By establishing a sequence of bridging distributions, AIS bypasses the rigid energy barriers of the glassy state. Furthermore, due to Jensen's Inequality, the logarithmic evaluation of the unbiased partition function estimator provides an accurate, strictly conservative *lower bound* for the conditional joint entropy within the 64×64 grid.

Joint Entropy Results. As illustrated in Figure 2, empirical AIS results reveal a robust linear relationship ($R^2 \geq 0.99$) between the defect count and the resulting boundary-conditioned joint entropy within the glassy phase of the state space. Both the 100% and 70% boundary constraint regressions yield a consistent, parallel slope of ≈ 5 bits per defect. This demonstrates that each localized edge constraint relaxation (a defect) uniformly expands the available state space by a factor of approximately $2^5 = 32$. The parallel nature of these curves indicates that partially relaxing the public key boundary induces a static, positive entropy offset but fundamentally does not alter the thermodynamic scaling mechanics of the interior grid. From a cryptographic perspective, this structural linearity is highly significant. It empirically demonstrates the absence of "entropy collapses," hidden structural hierarchies, or unpredictable phase transitions within the operational target band of 160–170 defects and the adjacent regions.

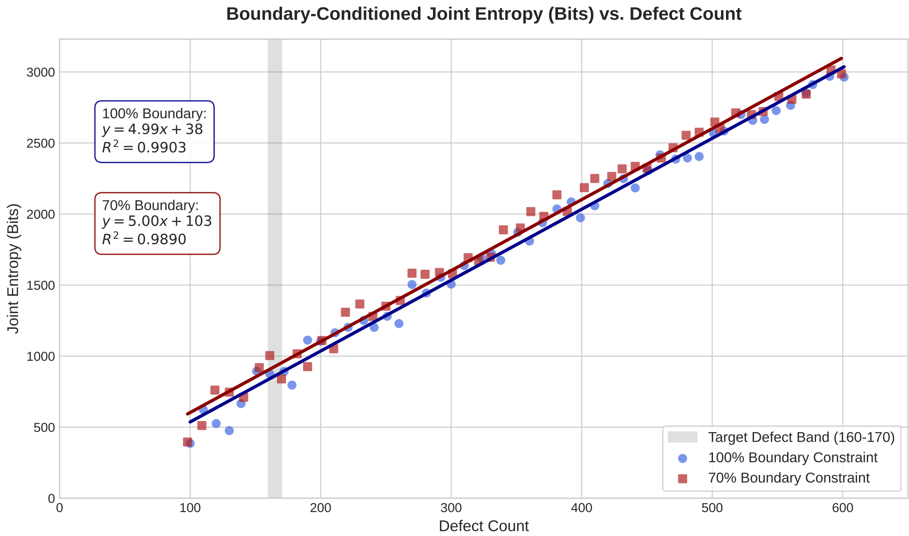

Figure 2: Boundary-Conditioned Joint Entropy vs. Defect Count. The operational glassy phase exhibits strict linearity ($R^2 \geq 0.99$). The parallel slope of ≈ 5 across distinct boundary constraint retention levels indicates that each defect reliably contributes 5 bits of entropy to the conditional state space.

Furthermore, the macroscopic phase transition is empirically corroborated by the AIS cooling schedule (Figure 3A). At $\beta \approx 2.44$, which corresponds to $D \approx 1250$, the system undergoes a rapid state collapse, after which the cooling curve asymptotically flattens. We verified this collapse by calculating the specific heat capacity ($C_v = \frac{\partial \langle E \rangle}{\partial T}$) curve (Figure 4). We observe a broad, rolling, Schottky-like peak, which is the hallmark of a material with geometric frustration. Instead of a divergent, λ-type peak demarcating a sharp phase transition, we observe a wider crossover regime where local constraints begin to dominate. The pronounced drop as cooling continues confirms that the annealing grid experiences a rapid loss of configurational degrees of freedom, locking into a topologically rigid glassy phase.

In summary, we empirically establish an absolute minimum joint entropy of 900 bits for QWTSS $\mathcal{SK}$ at the lowest boundary of the operational band ($D = 160$, $\rho_B = 0.70$). The specific annealing schedule of our key generation algorithm yields an empirical average of ≈ 940 bits of boundary-conditioned joint entropy across the operational band of D, at an effective average inverse temperature of $\beta \approx 4.5$. Importantly, the AIS methodology guarantees that these joint entropy estimates are strictly lower bounds.

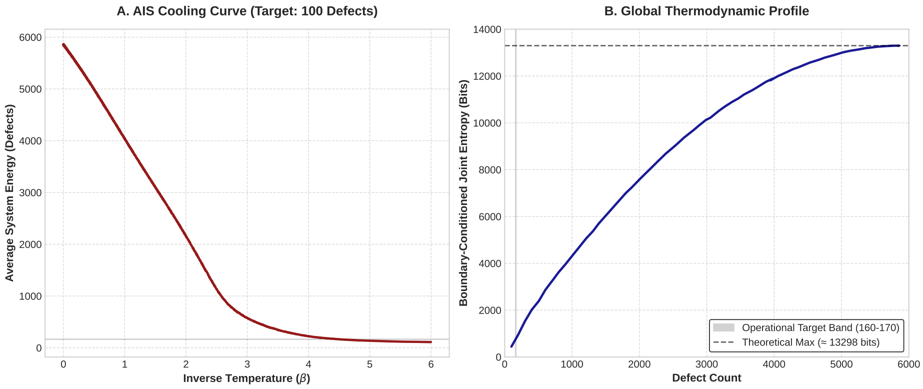

Figure 3: Global thermodynamic behavior of the QWTSS state space. **(A)** The AIS cooling schedule successfully thermalizes the system, smoothly bridging the high-energy liquid phase and the rigid glassy phase. **(B)** A macroscopic evaluation of the conditional joint entropy across all possible defect counts.

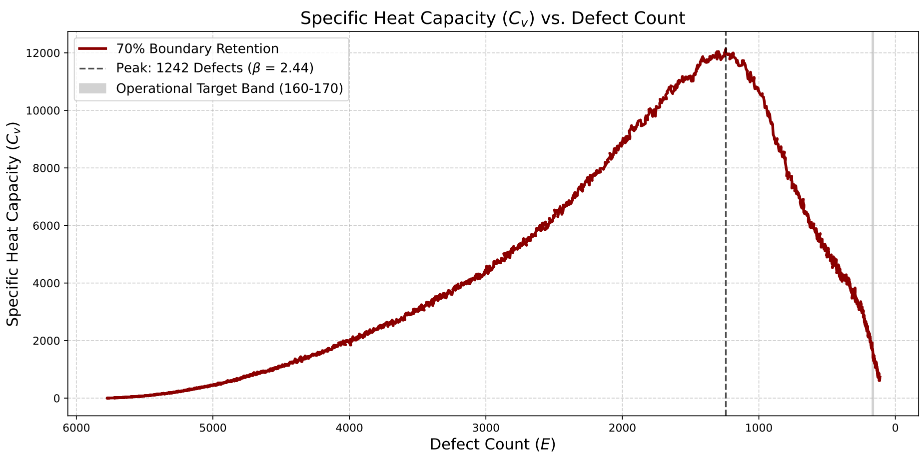

Figure 4: The global specific heat capacity curve shows hallmarks of a material with geometric frustration. The broad, rolling peak confirms a smooth crossover into the glassy phase.

4.2 Topological Analysis

While increasing the allowed defect count D predictably injects additional boundary-conditioned joint entropy, it also inevitably erodes the NP-hardness of the underlying problem if taken too far. If the grid accumulates too many unconstrained degrees of freedom, the long-range order of the aperiodic set collapses, allowing heuristic solvers to manipulate local regions of the grid without triggering cascading constraint violations. To better understand this upper constraint on D, we analyze three fundamental topological metrics of the state space.

By treating the tiling grid as an undirected graph $G = (V, E)$, where the vertices V represent the N tiles and the edges E represent perfectly matching tile borders (excluding defect edges), we can quantify the structural integrity of the lattice.

1. Giant Connected Component (GCC). The Giant Connected Component measures the fraction of vertices belonging to the largest connected subgraph of matching tiles. Mathematically, it is defined as:

$$\mathrm{GCC} = \frac{\max_{c \in C} |V_c|}{|V|} \tag{20}$$

where C is the set of all connected components in G.

Empirical analysis reveals that the GCC remains consistently at ≈ 1.0 across the entire operational defect band ($160 \leq D \leq 170$) and across all boundary retention levels. This demonstrates that the grid never breaks into isolated, disjoint regions of valid tiles. The system remains globally connected, meaning that defects are sparsely embedded within a singular, massive crystalline structure rather than physically fragmenting the grid. Even at $D = 600$, GCC > 0.999.

2. k-Core Rigidity (3-Core). While the GCC proves the grid is connected, it does not prove it is rigid. For this, we measure the 3-core of the graph. A k-core is defined as the maximal subgraph $H \subseteq G$ in which every vertex $v \in H$ has a degree $\deg_H(v) \geq k$. Because tiles in our planar grid have a maximum degree of 4, the 3-core represents the fraction of tiles that are structurally locked in by at least 3 correct matching edges, *recursively.*

The 3-core is an ideal topological signature of NP-hardness in Wang tiling, as it represents the extent of strong *recursive* coupling within the lattice. If a large 3-core exists, flipping a single tile to resolve a defect will almost always violate an existing matching edge, forcing a cascading chain of flips (an avalanche) that traps greedy solvers.

Our data shows that the 3-core is very robust in the deep glassy phase, remaining above 95% on average for $D < 400$. However, as D approaches 500, the 3-core undergoes an abrupt phase transition, dropping to $\approx 12\%$ at 600 defects. The 3-core variance also spikes concomitantly (Figure 5A). Similarly, for a bounded defect count of 160–170, relaxing the boundary pinning density (ρ_B) below $\approx 40\%$ retention triggers an identical structural collapse (Figure 5B). This demonstrates that pushing the system beyond the established operational limits risks entering a more liquid and trivial regime. While the 2-core metric (not shown) remains high near 1.0, the 2-core conceptually represents "bridges" between tiled regions, whereas the 3-core represents the "interlocking trusses."

3. Cyclomatic Density. Finally, we evaluate the density of topological loops within the matching grid. In graph theory, the cyclomatic complexity C measures the number of independent cycles in a planar graph, given by:

$$C = E - V + P \tag{21}$$

where P is the number of connected components. We normalize this value against the theoretical maximum of a defect-free grid to obtain the Cyclomatic Density.

Cycles represent redundant, interlocking constraints. A high cycle density implies that tiles are not merely forming linear chains, but are locked into closed planar loops. From an algebraic cryptanalysis perspective, C directly maps to the equation density of the constraint system; every independent cycle introduces a localized, circular chain of variables that an algebraic solver must satisfy simultaneously. This topological redundancy is what forces heuristic solvers to backtrack; attempting to push a defect along a path frequently results in the defect colliding with the other side of a closed loop. The empirical data demonstrates that the cyclomatic density decays smoothly and linearly as defects increase, confirming that the grid retains a high concentration of planar constraints within the 160–170 operational band.

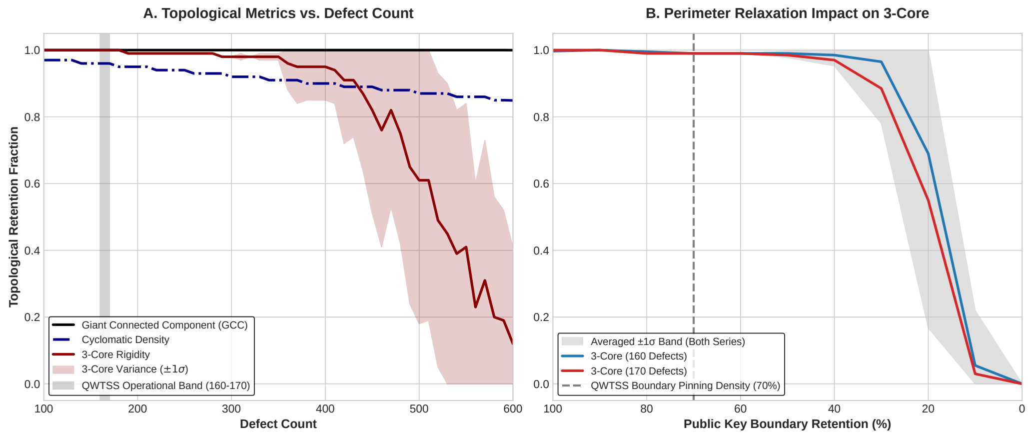

Figure 5: Topological metrics of the QWTSS state space. **(A)** The 3-core rigidity remains stable within the operational target band, but experiences a catastrophic phase collapse at $D > 400$. **(B)** At operational defect levels, the 3-core backbone also collapses when the boundary pinning density (ρ_B) is relaxed below a critical threshold of 40%.

5 State Space Characterization

5.1 Empirical Validation of Phase Criticality

As established in Section 2.3, the structural security of the QWTSS scheme relies on operating within the critically constrained, shattered phase of the solution space. To demonstrate that the operational parameters anchor the system in this intractable regime, we quantify the residual topological entropy per tile, denoted as ϵ.

In the thermodynamic limit of CSPs, the effective global branching factor b_{eff} governs the survival rate of global search paths across a lattice of size N. It is defined relative to the total number of valid global states $\langle\Omega\rangle$:

$$b_{\text{eff}} = \langle\Omega\rangle^{\frac{1}{N}} \tag{22}$$

We can express this branching factor as $b_{\text{eff}} = e^{\epsilon}$, where ϵ is the residual topological entropy measured in nats. By taking the natural logarithm, we derive the formal definition of ϵ:

$$\epsilon = \frac{1}{N}\ln\langle\Omega\rangle \tag{23}$$

To compute $\langle\Omega\rangle$ for the QWTSS lattice, we rely on our earlier AIS estimations. Unlike localized branching calculations—which fail to account for the geometric cycle-closures inherent to 2D topological boundaries—the AIS chain natively enforces all stringent edge-matching rules and macroscopic boundary conditions over the entire $N = 4096$ lattice.

Our AIS estimation yields a boundary-conditioned global joint entropy of $H \approx 940$ bits for the operational regime: $D \in [160, 170]$ and $\rho_B = 0.70$. The total valid state space is therefore

$\langle \Omega \rangle \approx 2^{940}$. Converting this bit-entropy into nats to solve for ϵ:

$$\epsilon = \frac{H \ln 2}{N} \approx \frac{940 \times 0.693}{4096} \approx 0.159 \text{ nats} \tag{24}$$

Because $\epsilon \approx 0.159 > 0$, the system is mathematically guaranteed to be satisfiable (SAT), ensuring that valid private keys ($\mathcal{SK}$) can be generated by users. The corresponding global effective branching factor is:

$$b_{\text{eff}} = e^{0.159} \approx 1.172 \text{ tiles per coordinate} \tag{25}$$

Because the raw aperiodic alphabet contains significantly more potential states ($|\mathcal{A}| = 11$ for the Jeandel-Rao set), this severely suppressed branching factor reveals a considerable level of mutual information across the grid. The local edge-matching constraints force the tile distributions to be highly interdependent; successfully resolving the state of a local region heavily dictates the valid choices of its neighbors, which is a primary hallmark of geometric frustration.

Crucially, this low-positive entropy density is the exact mathematical signature of the critically constrained regime. The severely suppressed value of $\epsilon \approx 0.159$ confirms that the macroscopic solution space is exponentially sparse. While valid states exist, they represent a vanishingly small fraction of the total configuration space. This extreme scarcity forces exact CDCL SAT and SMT solvers into maximum backtracking depth; they must search prohibitively deep into the decision tree before discovering these rare valid configurations or proving localized contradictions.

5.2 Empirical Demonstration of the Overlap Gap

To characterize the topological isolation between valid QWTSS solution clusters—the Overlap Gap—we employ a three-pronged empirical approach: statistical limit analysis via Extreme Value Theory (EVT), thermodynamic barrier profiling, and macroscopic geometric derivation.

The Asymptotic Floor (EVT). We first executed several multi-day runs of streaming nearest-neighbor searches, tracking the progression of the global minimum Hamming distance, $HD_{min}(N)$, between independent $\mathcal{SK}$ pairs (all conditioned on a single $\mathcal{PK}_{master}$ for a given run) as a function of total pairwise comparisons (N). As shown in the logarithmic plot of Figure 6, the decreasing nearest-neighbor distances monotonically flatten at increasing depths, clearly violating an unbounded Gaussian assumption and demonstrating a hard topological floor.

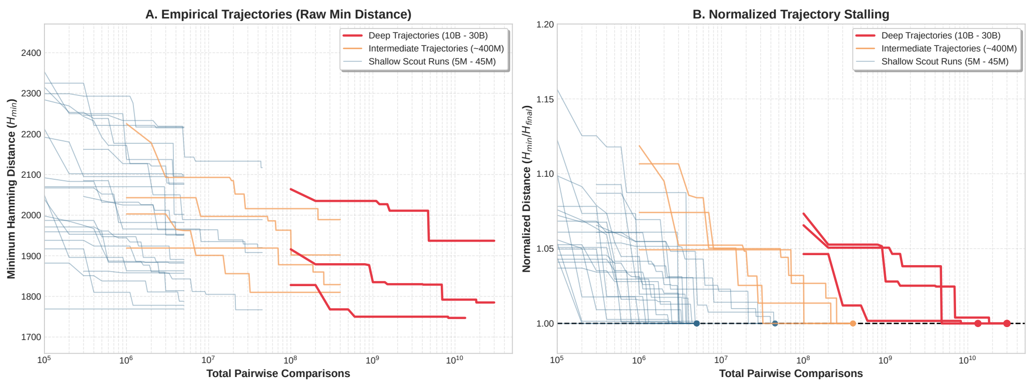

Figure 6: Global minimum Hamming distance trajectories. **(A)** The raw minimum Hamming distance ($HD_{min}(N)$) as a function of total pairwise comparisons (N) observed across independent trajectories of $\mathcal{PK}$-conditioned $\mathcal{SK}$ pairs. The decreasing nearest-neighbor distances stall out asymptotically at the topological barrier (Δ_{OGP}). **(B)** The normalized convergence ($HD_{min}(N)/HD_{min}(final)$) isolates algorithmic search efficiency from initial conditions.

According to the Fisher-Tippett-Gnedenko theorem, such bounded extreme minima must converge to a Type III (Weibull) extreme value distribution. By applying a Non-linear Least Squares (NLS) optimizer to the empirical extreme value tails of our three deepest runs (detailed in Appendix C), we calculated the terminal asymptotes representing the absolute minimum distance (Δ_{OGP}) separating any two disjoint clusters for each run. These asymptotes resolved with good statistical confidence ($0.77 \le R^2 \le 0.87$), yielding a pooled weighted estimate of $\Delta_{OGP} \approx 1709$ tiles (Figure 7). While the exact Δ_{OGP} estimates vary slightly for the different $\mathcal{PK}$ trajectories, these charts clearly illustrate that they are tightly clustered.

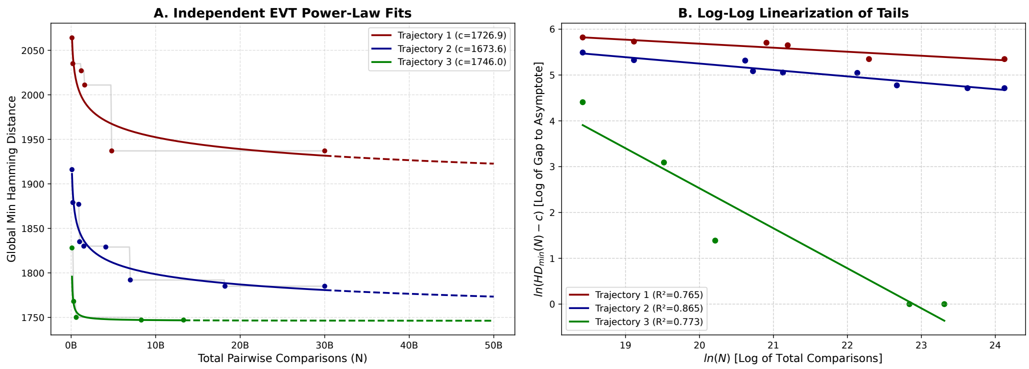

Figure 7: EVT asymptotic estimation of the Overlap Gap. **(A)** The power-law fits of the three deepest runs in linear space. **(B)** The log-log coordinate linearization of the extreme value tails used to optimize the regressions.

This theoretical minimum distance holds strictly for keys generated by separate, uncorrelated random annealing trajectories. While local solution clusters containing minor permutation differences do exist, the probability of independently sampling two keys from the same cluster is statistically zero. Consequently, in aggregate, the pooled global mean Hamming distance between independent keys is $\mu = 3055$ ($\sigma = 275$) tiles, indicating that independently annealed keys are highly orthogonal (naturally differing across $\approx 75\%$ of the grid). Furthermore, the $\Delta_{OGP} \approx 1709$ asymptotic barrier dictates that the probability of randomly generating an $\mathcal{SK}$ that shares more than 58% of its spatial structure with an independent target $\mathcal{SK}$ is effectively zero.

The Thermodynamic Barrier. To understand the energy landscape enforcing this gap, we analyzed the transitional state space between the closest discovered ($\mathcal{SK}_A, \mathcal{SK}_B$) pairs. We generated thousands of $\mathcal{SK}_A \to \mathcal{SK}_B$ transition trajectories using a specialized Stochastic Flat-Path Greedy Solver (SFPGS). To map the path of absolute least resistance, the SFPGS was artificially gifted with two omniscience advantages: *Target Omniscience* (restricting tile flips to only those that monotonically approach $\mathcal{SK}_B$) and *Selection Omniscience* (evaluating the entire local neighborhood to prioritize "flat" isentropic moves over climbing the energy gradient).

Despite these artificial advantages, our 1-to-1 stochastic path density histograms (Figure 8) and 1-to-N radial mapping (Figure 9) visually confirm that continuous, easily traversable manifolds between valid keys do not exist. We specify the exact experimental procedure and visualization details in Appendix D. After the SFPGS solver exhausts the available isentropic permutations (the initial bright, narrow line), it is invariably forced into a rugged energy landscape (the diffuse band, typically emerging around $t_{norm} \approx 0.2$). We define the *Activation Energy Barrier* (ΔE_{act}) as the peak thermodynamic excursion required to cross this dead zone. The SFPGS-optimized trajectories revealed an absolute minimum excursion of $\Delta E^*_{act} \approx 20$ defects, with an average of $\Delta E_{act} \approx 42.7$.

By evaluating the standard Metropolis-Hastings acceptance probability at the operational

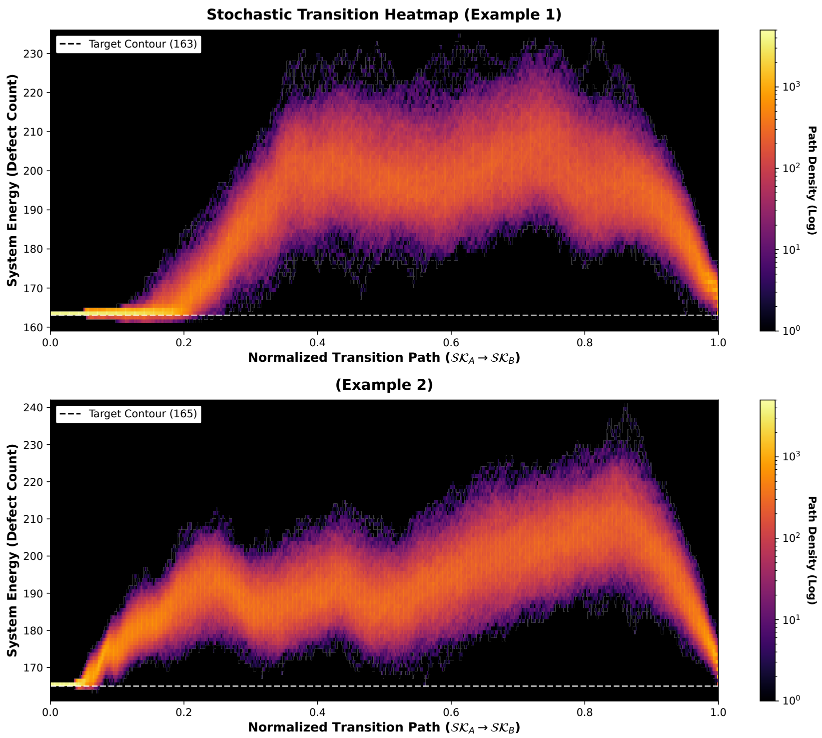

Figure 8: A logarithmic path density heatmap mapping 1000 stochastic transition walks between a single $\mathcal{SK}$ pair (each example uses a separate pair). The fragmentation of these near-optimal SFPGS paths highlights the lack of continuous isentropic manifolds.

inverse temperature ($\beta = 4.5$), the maximum probability of a constant-temperature MCMC solver surmounting this empirically observed barrier is effectively zero, even with both omniscience advantages:

$$P_{climb}\Big|_{\text{SFPGS}} = \exp(-\beta\Delta E^{*}_{act}) = \exp(-4.5 \times 20) \approx 8.2 \times 10^{-40} \tag{26}$$

Because Boltzmann weights are path-independent, partitioning this climb across intermediate states provides no advantage. Furthermore, if we remove the SFPGS's lookahead advantage to evaluate "blind" shortest Hamming paths, the average peak defect count reaches $D \approx 1290$. This exceeds the macroscopic glass transition threshold ($D \approx 1250$), indicating that traversing the naïve shortest paths requires a complete phase re-liquefaction, offering no computational shortcut over a brute-force restart from $T = \infty$.

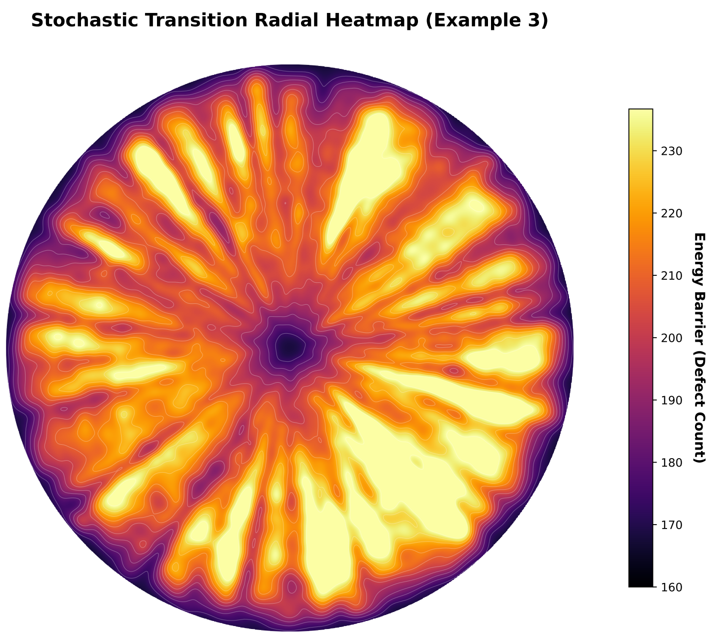

Figure 9: A 2D radial topographical heatmap mapping 100 independent SFPGS-optimal transitions from a central $\mathcal{SK}_{master}$ solution cluster to unique perimeter $\mathcal{SK}$ targets sharing the same boundary constraints in nearby solution clusters (randomly ordered).

Geometric Derivation of the Overlap Gap. Finally, the SFPGS trajectory data allow us to derive the Overlap Gap geometry. During the initial transition phase, the SFPGS navigates the solution cluster's internal isentropic manifold without inducing significant energy penalties. Over the aggregate dataset, this targeted flat-path navigation persists for an average of 21.8% of the normalized transition distance before first breaching the operational band's upper bound ($D = 170$). By anchoring this $t_{norm} \approx 0.218$ threshold to the expected global Hamming distance between independent keys ($\mu = 3055$), we can estimate the maximum radial extent, R^*, of a single valid cluster:

$$R^* \approx t_{norm} \times \mu \approx 0.218 \times 3055 \approx 666 \text{ tiles} \tag{27}$$

If we select two random, valid solution clusters separated by the expected distance μ, the gap G between their closest topological edges is the total distance minus their respective maximum radial extents:

$$G = \mu - 2R^* = 3055 - 2(666) \approx 1723 \text{ tiles} \tag{28}$$

This macroscopic geometric derivation of $G \approx 1723$ tiles perfectly aligns with the theoretical lower bound estimate of $\Delta_{OGP} \approx 1709$ tiles derived via our independent EVT analysis. This convergence of statistical, thermodynamic, and geometric data provides compelling evidence of the Overlap Gap's presence in the QWTSS operational band.

5.3 Solution Cluster Characterization

Cluster Volume Estimation. Obtaining realistic estimates of a given cluster's total entropy proved quite difficult; even specialized AIS variants failed to produce reliable data in this regard. Ultimately, we found success using a modified form of Monte Carlo integration based on the Rosenbluth-Knuth estimation method [38, 20]. While this technique was originally designed to estimate the size of self-avoiding walks and combinatorial search trees via stochastic depth-first probes, we adapt it into a geometric "ray-casting" approach. Rays are traced outward stochastically from an anchor $\mathcal{SK}_i$ to map the surrounding state space. Crucially, we introduce a dynamic commutative correction to adapt the estimator from a strict tree topology to the layered Directed Acyclic Graph (DAG) structure of the tiling permutations (Appendix E). Finally, we address the anchor centrality bias by generating an ensemble of N anchors via a long-range MCMC diffusion walk within the cluster. The final cluster volume estimate is defined as the maximum value captured across the ensemble: $V_{\text{est}} = \max_{0 \le i < N}\{V(\mathcal{SK}_i)\}$, where $V(\cdot)$ denotes the corrected ray-casting volume estimator.

We applied this methodology across 30 independent clusters, generating an $N = 10$ intra-cluster anchor ensemble for each cluster and casting $M = 30,000$ corrected rays per individual anchor. The pooled convergence curve (Figure 10A) demonstrates stable convergence of the running LogSumExp volume estimator $V(\cdot)$, confirming that our ray count was sufficient to accurately capture the locally accessible volume from each anchor location. The smallest cluster sampled was 84 bits, while the largest was 193 bits, revealing a substantial 33 decimal orders of magnitude between the extrema.

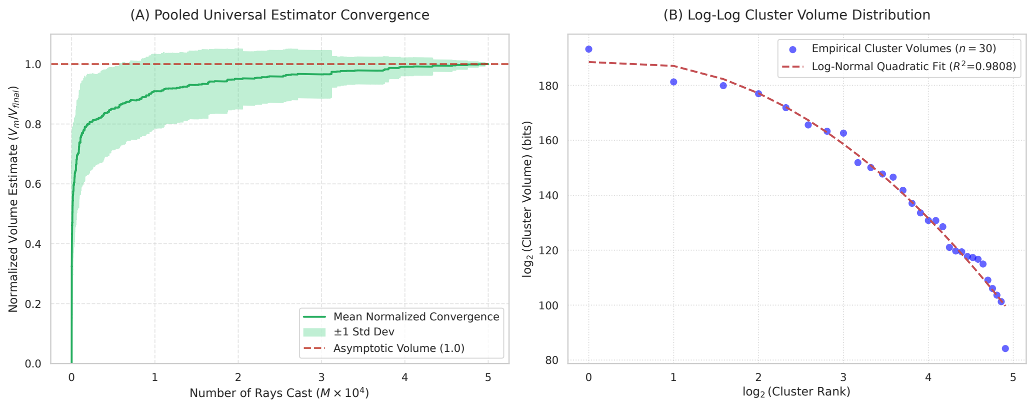

Figure 10: (A) The pooled estimator convergence curve (cumulative LogSumExp average), normalized and averaged across 50 independent exploratory anchor samples ($M = 50,000$). **(B)** The rank-size distribution of the $n = 30$ sampled cluster volumes in log-log space. The tight quadratic fit ($R^2 = 0.98$) indicates a log-normal scaling, implying a multiplicative cascade process.

Cluster Invariant Property I: Multiplicative Cascade. Because combinatorial phase space probability is strictly proportional to entropic volume ($V = 2^H$), large clusters act as potent probabilistic attractors. Due to the aggressive nature of exponential scaling, the combinatorial mass of our $n = 30$ sampled cluster ensemble is overwhelmingly concentrated in the single largest extremum. Specifically, Cluster 19 ($2^{193.19}$ states) accounts for exactly 99.963% of the ensemble's entire combined sample volume ($V_{\text{combined}} = 2^{193.1905}$). Accordingly, the empirical discovery of *relatively* microscopic clusters ($H \approx 84$ bits) alongside a giant extremum ($H \approx 193$ bits) via the uniform random sampling of $\mathcal{SK}_0$ starting anchors presents a volumetric paradox that can only be resolved by the density of states.

For an 84-bit cluster to be sampled with a similar frequency as a 193-bit cluster—despite possessing 2^{-109} of the probabilistic cross-section—the underlying multiplicity of the phase space must increase exponentially to compensate. A vast multitude of mid-to-small volume clusters effectively shatters the phase space, collectively balancing the probabilistic weight of the rare "mega-clusters." While a first-order power law approximation yields $R^2 = 0.90$, the cluster volume distribution is more accurately characterized by a log-normal scaling ($R^2 = 0.98$), as shown in Figure 10B. Accordingly, despite the combinatorial dynamic of the state space, the QWTSS solution cluster's expected logarithmic volume is only ≈ 137.5 bits (the median of the log-normal distribution).

This parabolic curvature in log-log space is highly suggestive of multiplicative attrition. It implies that the solution space is governed by a multiplicative cascade, where the entropic capacity is determined by the stochastic composition of local constraints across the cluster manifold. The multiplicative cascade is the hallmark of a stochastic multifractal, a standard model for the phase space of structurally frustrated lattices [34]. The extreme computational complexity of aperiodic Wang tilings manifests as inescapable topological contradiction; the accumulation of the associated constraints shatters the QWTSS phase space into a hierarchical, "glassy" landscape [30, 11].

Uniquely Isolated Clusters. To estimate the number of distinct solution clusters $N_{clusters}$ comprising the total solution volume $V_{total} = 2^{940}$, we must account for the volume-dependent multiplicity. By integrating the volume-weighted frequency distribution to the thermodynamic limit (Appendix F), we derive an estimated total count of uniquely isolated QWTSS solution clusters:

$$N_{clusters} \approx 2^{848} \tag{29}$$

Despite this large figure, these discrete solution clusters are embedded within a raw lattice configuration space that exponentially dwarfs them ($11^{4096} \approx 2^{14,170}$). Consequently, the landscape of valid configurations remains both extremely sparse and fragmented.

Cluster Invariant Property II: Severe Anisotropy. Another finding from the ray-casting analysis is that the mass contribution of any given solution cluster is completely dominated by a singular topological direction. This extreme Pareto distribution is evident in Figure 11B, which plots the proportion of total cluster volume contributed by the top 10 rays, according to their rank within their individual clusters. Notably, out of $M = 30,000$ rays cast per cluster, the single top-ranked ray contributes $\approx 93.0\%$ of that cluster's total entropic volume on average, while the top two rays combined account for over 98.6% of the valid $\mathcal{SK}$ content. Rather than an isotropic globular manifold, the data indicate that a QWTSS valid-key cluster forms a narrow "hyper-needle" in Hamming space. Corresponding to the top-ranked ray, the primary axis usually dominates the combinatorial volume and indicates a severely skewed thermodynamic topology within the hypercube.

This severe anisotropy also explains an empirical asymmetry observed in our earlier SFPGS-optimized normalized transition trajectories (Figure 8): the continuous, isentropic flat-path navigation regime appears exclusively at the *start* of the transition ($\mathcal{SK}_A$), but never at the termination ($\mathcal{SK}_B$). When departing the origin, the greedy SFPGS naturally discovers and traverses the cluster's dominant principal axis, allowing it to reliably increase the Hamming distance without incurring any significant energy penalty. However, once the solver crosses the rugged Overlap Gap, its final descent into the target cluster $\mathcal{SK}_B$ is invariably steep and abrupt. In high-dimensional Hamming space, a random approach vector arriving from the energetic void

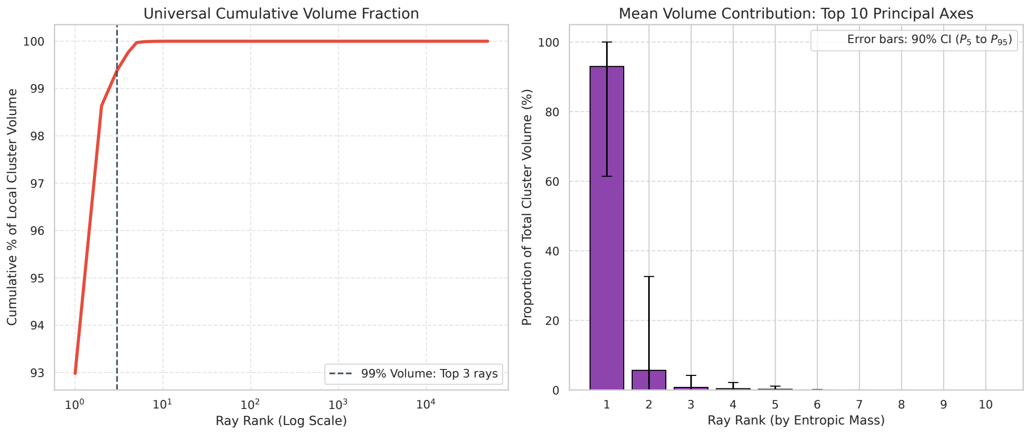

Figure 11: Cluster anisotropy (aggregated data). **(A)** The cumulative volume contribution curve in logarithmic scale illustrates that the solution clusters are highly degenerate, with 99.99% of the minor axes providing essentially zero entropy. **(B)** The extreme Pareto distribution, wherein the single principal axis alone accounts for ≈ 93% of the cluster's entire valid $\mathcal{SK}$ content on average.

is overwhelmingly likely to be orthogonal to the singular principal axis of the $\mathcal{SK}_B$ solution cluster. Consequently, the solver is forced to enter through one of the 99.99% of minor axes that possess a negligibly small radial extent.

Cluster Invariant Property III: Dendritic Tapering. To understand the pressures shaping the boundary of a solution cluster, we investigated the marginal entropy impacting each ray at each step within the ray-caster routine. By analyzing this curve as a function of the ray penetration depth normalized by its terminal length ($\hat{N}$), we discover another invariant property. The macroscopic expectation of these trajectories, across every anchor in every sampled cluster, follows a consistent pattern characterized by two distinct phases (Figure 12):

1. **The Origin Cliff:** At the very beginning of the ray-cast, the decay rate is nearly vertical. Before the first permutation, the global grid state ($\mathcal{SK}_{master}$) averages $\approx 36,700$ valid tile swaps across the board, utilizing the initial unspent defect budget (e.g., within the $160 \leq D \leq 170$ operational band, an $\mathcal{SK}_{master}$ starting at $D = 165$ possesses 5 free defects). However, as this limited defect budget is spent, the average available moves plummet to $\approx 14,300$ after the first tile flip, $\approx 1,850$ after the second, and just 480 by the third. This sharp cliff locks the initially selected pathway into highly constrained corridors almost immediately.

2. **The Long Decay:** From the fourth step onward, the trajectory enters a gradual but intensifying period of geometric starvation. Each step outward shaves off remaining degrees of freedom. Approximately halfway through the average ray's normalized lifetime, the marginal entropy turns permanently negative, actively contracting the cluster manifold. These compounding 2D spatial constraints squeeze the permutation trajectories into mere filaments, invariably ending in geometric checkmate.

Thus, in every single direction, a unique topological funnel exists, squeezing the global commutative frontier down to a narrow channel before it ultimately terminates at zero degrees of freedom. This reveals the true multifractal nature of the cluster's manifold, exhibiting severe dendritic tapering along every outward branch and across *all* Hamming length scales. Crucially, as the marginal entropy curve drops below 1.0 bit/step, the permutation trajectories are forced into a regime of sub-binary determinism. Because a discrete grid step must offer at least one valid tile

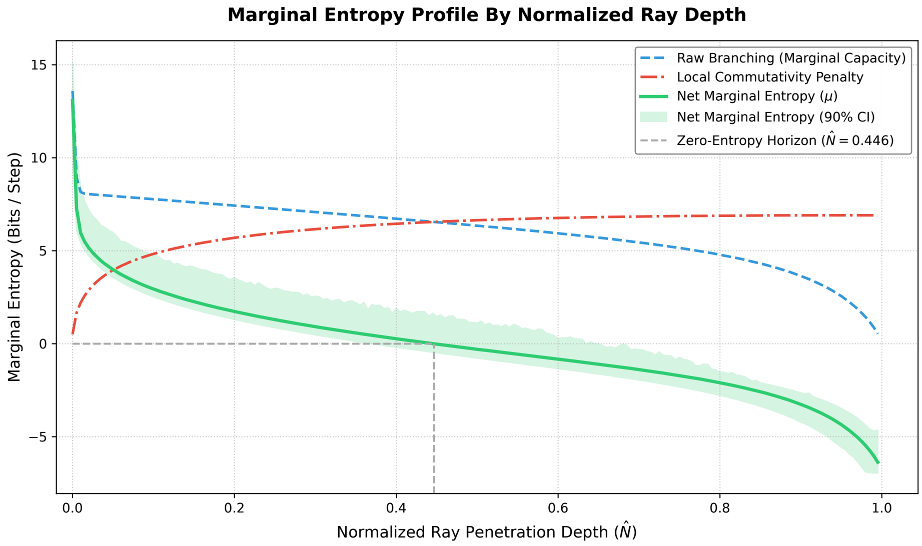

Figure 12: Global aggregate of the solution cluster marginal entropy profile as a function of normalized ray penetration depth ($\hat{N}$). The raw branching capacity (blue dashed line) represents the uncorrected marginal geometric degrees of freedom at each depth. This capacity is penalized by the local commutativity estimate (red dash-dot line) to yield the net marginal entropy μ (solid green line). The shaded green region denotes the 90% confidence interval of μ across the ensemble. The zero-entropy horizon (gray dashed line) marks the critical depth where the manifold transitions from expansion into contraction ($\mu < 0$). All series represent a pooled average over 50 evaluated clusters.

swap, a local average below 1.0 bit statistically necessitates the presence of tight, and increasingly frequent, zero-entropy bottlenecks ($\log_2(1) = 0$ bits). This produces purely deterministic, non-branching sequences of varying lengths that increasingly dominate the path. The net effect of this geometric starvation is clearly illustrated in the multi-threshold survival analysis of Figure 13.

This progressively negative marginal entropy profile is a universal dynamic directly stemming from the severe geometric frustration of the JR-11 grammar. Consequently, a solution cluster's maximum entropic volume is not dictated by any hypothetical localized variances in grammatical freedom, but strictly by the radial distance the manifold can sustain against this universal topological pinch-off. Larger clusters simply *survive longer* due to greater central defect budgets or serendipitous tiling alignments. Fundamentally, because a single 1D Hamming step (one tile swap) immediately imposes up to four 2D spatial edge constraints, this geometric friction inevitably outpaces the manifold's expansion. These 2D constraints rigidly percolate outward, choking off entire sub-trees of commutative options and guaranteeing the ultimate termination of every pathway—even those aligned along a cluster's principal permutative axis. We posit that this universal and inescapable geometric starvation is the mathematical origin of the shattered QWTSS state space.

Solution Cluster Visualization. To visualize the high-dimensional phase space of a valid solution cluster, we developed an empirically coupled 3D topological manifold. This manifold was derived from the 250 rays exhibiting the greatest penetration depths, utilizing the pairwise Hamming distances between their terminal points. The macroscopic structural geometry was extracted using Torgerson Metric Multidimensional Scaling (Classic MDS). By performing an eigendecomposition on the pairwise Hamming distance matrix—augmented to include the Step 0 origin anchor—we mapped the terminal boundary constraints into $\mathbb{R}^3$ while preserving the cluster's principal anisotropic axes. The internal dendritic topology was then reconstructed via

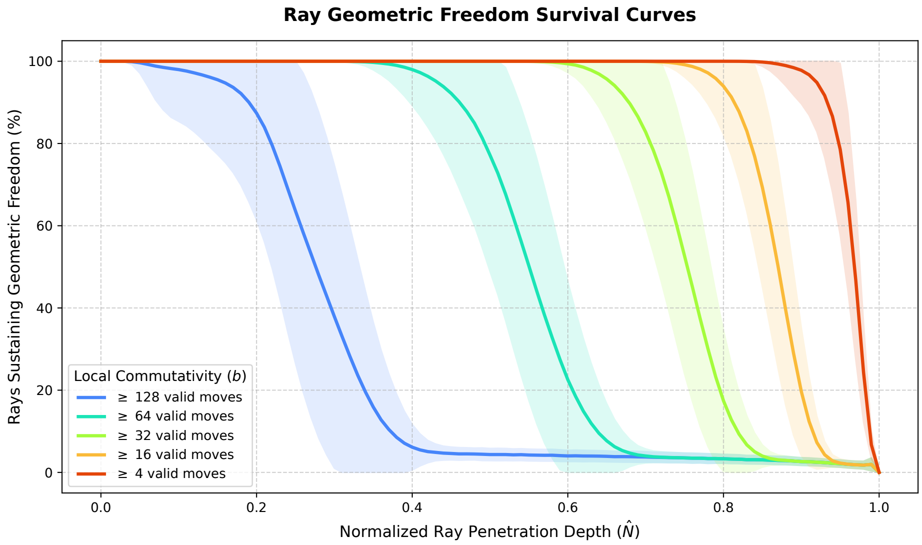

Figure 13: Global ensemble survival curves demonstrating the decay of geometric freedom as a function of normalized ray penetration depth ($\hat{N}$). Each solid trajectory tracks the mean percentage of evaluation rays capable of sustaining a local commutativity branching factor (b) at or above a specified threshold. The correspondingly colored shaded bands represent the 1σ inter-cluster variance across the evaluated ensemble ($n = 50$). The accelerated collapse of these survival probabilities empirically captures the universal topological funnel encountered by every ray.

agglomerative hierarchical clustering (Ward's minimum variance method), yielding a bifurcating graph of the major combinatorial branching pathways discovered by the rays.

To accurately model the thermodynamic capacity of each individual branch, we utilized VTK-based volumetric rendering to physically map the empirical local marginal entropy to the cylindrical volume of the branches. The ray's normalized penetration depth $\hat{N}$ was linearly interpolated along each topological edge to continuously query the empirical marginal entropy profile $\mu(\hat{N})$. To ensure that the rendered 3D cross-sectional area remains directly proportional to the true local entropic volume ($V \propto 2^{\mu}$), the cylindrical radius r was mapped to the square root of the spatial branching capacity ($\sqrt{2^{\mu(\hat{N})}}$). Furthermore, to prevent the initial combinatorial explosion at the origin from visually eclipsing the bases of the topological funnels in the cluster's center, the spatial mapping applies a lower-bound depth cutoff at $\hat{N} \geq 0.01$ (Figure 14). Finally, by truncating the rendering to only the 250 deepest combinatorial branches, this visualization explicitly omits the fractal dendritic outgrowths occurring within the manifold at all smaller scales, as the resulting visual noise would severely occlude the macroscopic structure.

5.4 Summary

Together, these results confirm that the $\mathcal{PK}$-conditioned state space of valid $\mathcal{SK}$ solutions is a vast archipelago of roughly 2^{848} distinct, disjoint, and isolated cryptographic "islands" inhabiting the critically constrained, shattered phase. Structurally, each solution cluster manifests as a highly degenerate hyper-needle dominated by a single permutative axis, with multifractal dendrites tapering into the energetic void until they terminate in geometric checkmate. Because each island is separated by the Overlap Gap, no local search algorithm or Markov Chain solver can bridge the intervening space. Even stochastic solvers employing temporary thermal excitation (re-heating) would be forced to inject so much entropy to crest the interjacent barriers that the system effectively decorrelates, reducing the attempted traversal to a random, brute-force restart.

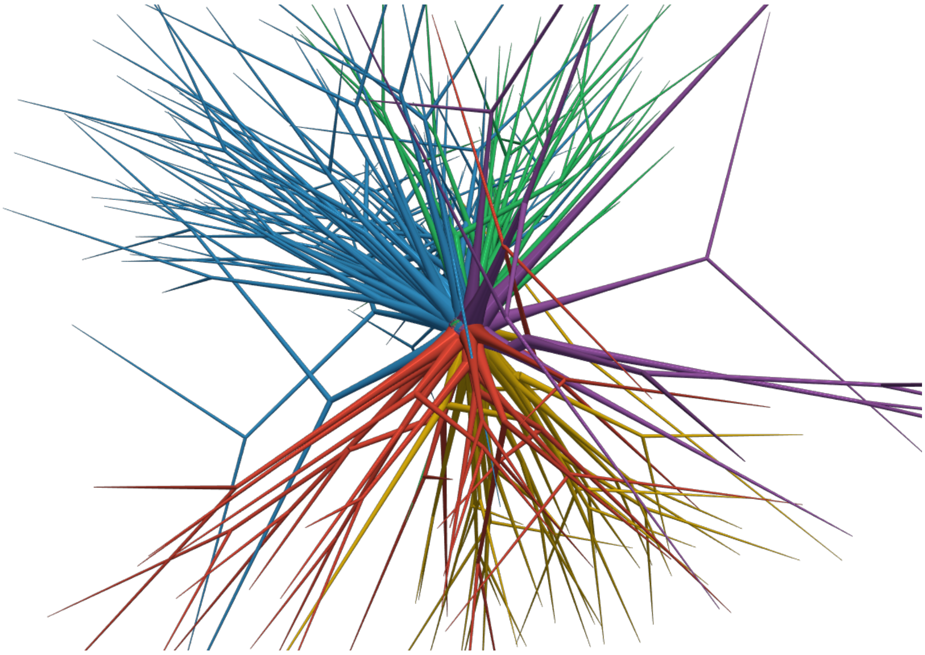

Figure 14: A rendering of a randomly selected cluster's macroscopic phase-space structure projected to 3D. The rendered cross-sectional volume accurately depicts the local marginal entropic capacity (2^{μ}) at every location where $\hat{N} \geq 1\%$. The thinning dendrites represent the starvation of geometric degrees of freedom as solution permutation corridors branch out from the higher-entropy core region. The colors represent common branch ancestry.

Formal Validation of the OGP Criteria. The empirical evidence confirms that the QWTSS scheme meets all three requirements for the OGP, as formalized by Gamarnik [13]:

1. **Near-Optimality** (μ-Optimality): The gap must exist for all solutions that achieve an objective value within a certain threshold of the absolute optimum.
   - The QWTSS scheme's operational band bounds valid $\mathcal{SK}$ configurations to a narrow thermodynamic threshold ($D \in [160, 170]$).
2. **Distance Discontinuity**: There exist two distances $0 < d_1 < d_2$ such that for any two near-optimal solutions x_1 and x_2, their distance is either $\leq d_1$ (they are in the same cluster) or $\geq d_2$ (they are in different clusters). The interval (d_1, d_2) is empty.
   - The EVT asymptote analysis established the absolute minimum inter-cluster gap (d_2) at $\Delta_{OGP} \approx 1709$ tiles. The Energy Barrier data tightly confirmed this d_2 value at $G \approx 1723$ tiles and also established the solution cluster maximum radial extent (d_1) at $R^* \approx 666$ tiles. The Hamming distances between ≈ 666 and ≈ 1709 are empty of valid $\mathcal{SK}$ configurations.
3. **With High Probability**: The property must hold with high probability (w.h.p.) over the randomness of the problem instance.
   - In aggregate, the empirical data confirm that this topological isolation persists universally across all experiments, sampled keys, and evaluated trajectories.

The empirically verified presence of the OGP implies the very high likelihood that QWTSS exhibits *average-case* intractability. All our characterizations of the QWTSS state space indicate a remarkably hostile landscape for many classes of potential solvers. Topologically, the search problem is akin to locating hyper-needles within an exponentially sparse hyper-haystack. Of note, the OGP has been formally proven to defeat quantum optimization algorithms, including Quantum Approximate Optimization Algorithms (QAOA) and quantum annealing [12]. Furthermore, the extreme combinatorial anisotropy of the QWTSS clusters acts as an independent defense multiplier against global quantum tunneling via exponential suppression of the quantum overlap integral.

Quantum Overlap Suppression via Extreme Anisotropy. To quantify this suppression, we analyze the quantum overlap integral. Specifically, we construct a comparative model to demonstrate how the severe configuration-space anisotropy of the QWTSS clusters fundamentally degrades tunneling fidelity when compared to isotropic clusters of the same thermodynamic volume.

Let the system be defined in a Hilbert space of dimension q^N, spanned by the basis states of the global discrete configuration space Σ^N. We define two theoretical target clusters, C_{iso} (a perfectly isotropic Hamming sphere) and C_{aniso} (the highly degenerate, k-dimensional hyper-needle observed in QWTSS). To isolate the structural impact of the topology, we constrain both clusters to possess the exact same combinatorial volume: $|C_{iso}| = |C_{aniso}| = V$.

By the definition of the Overlap Gap Property (OGP), both clusters are separated from the initialization state by a strict minimum Hamming distance barrier, Δ_{OGP}. Thus, both manifolds originate at boundary distance $d = \Delta_{OGP}$ and extend outward.

Let $\alpha(d)$ represent the probability amplitude of the tunneling wavefunction at distance d from the source. Due to the local nature of the driving Hamiltonian, the tunneling amplitude decays monotonically and rapidly as distance increases ($\alpha(d) \gg \alpha(d+1)$). The tunneling fidelity $\mathcal{F}$ for a uniform target state $|\psi_C\rangle$ is the squared sum of these amplitudes over the geometry of the cluster:

$$\mathcal{F}_C = \left| \frac{1}{\sqrt{V}} \sum_{x \in C} \alpha(d(x)) \right|^2 = \frac{1}{V} \left| \sum_{d=\Delta_{OGP}}^{\Delta_{max}} g_C(d)\alpha(d) \right|^2 \tag{30}$$

where $g_C(d)$ is the density of states of the cluster at distance d, subject to the conservation of volume constraint $\sum g_C(d) = V$.

For the isotropic cluster C_{iso}, the density of states $g_{iso}(d)$ scales combinatorially utilizing all N dimensions of the global lattice. Consequently, C_{iso} achieves the target volume V within a remarkably shallow maximum depth, R_{iso}. It effectively packs the entirety of its volume directly against the Δ_{OGP} boundary, maximizing its presence in the region where $\alpha(d)$ is strongest.

Conversely, the anisotropic cluster C_{aniso} is geometrically constrained to a few k principal axes ($k \ll N$). Because its lateral expansion is severely bottlenecked, its density of states $g_{aniso}(d)$ scales at a markedly reduced rate. To encapsulate the exact same volume V, the mass of C_{aniso} must distribute along a few narrow spines, pushing its effective thermodynamic depth to $R_{aniso} \gg R_{iso}$.

By the concentration of measure in a high-dimensional discrete sequence space Σ^N equipped with the Hamming metric, the internal k-dimensional sparse basis of C_{aniso} is overwhelmingly orthogonal to the displacement vector separating the statistically independent source and target clusters. Thus, C_{aniso} is forced to distribute the vast majority of its V states deeper into the configuration space, where the tunneling amplitude $\alpha(d)$ is exponentially weaker and its weighted sum is strictly constrained:

$$\left| \sum_{d=\Delta_{OGP}}^{\Delta_{OGP}+R_{aniso}} g_{aniso}(d)\alpha(d) \right| \ll \left| \sum_{d=\Delta_{OGP}}^{\Delta_{OGP}+R_{iso}} g_{iso}(d)\alpha(d) \right| \tag{31}$$

Therefore, for two clusters of identical thermodynamic volume situated behind an identical Overlap Gap barrier, the tunneling fidelity is severely suppressed by the anisotropic topology:

$$\mathcal{F}_{aniso} \ll \mathcal{F}_{iso} \tag{32}$$

In conclusion, the extreme combinatorial anisotropy of the QWTSS solution clusters acts as a fundamental and independent defense mechanism against global quantum tunneling. By forcing the manifold's thermodynamic mass to stretch away from the Overlap Gap boundary and into regions of the configuration space where the tunneling amplitude exponentially decays, this hyper-anisotropy severely suppresses the probability of successful quantum tunneling. Because this configuration-space geometry deprives the target of a broad isotropic neighborhood, it directly parallels the Anderson localization failure mode established by Altshuler et al. [2], which results in exponentially small spectral gaps that negate the quantum algorithmic speedup.

6 Implementation Details and Performance

To evaluate the practical viability of the QWTSS architecture, we implemented a complete reference pipeline in C++17[2]. The cryptographic core utilizes the StarkWare Stone Prover for the zk-STARK backend, while the MCMC tiling solver features dual execution pathways: a parallelized CUDA implementation for GPU acceleration, and a standard multithreaded implementation for CPU execution. Specific algorithmic and architectural optimizations deployed to accelerate $\mathcal{SK}$ convergence are detailed in Appendix G.

All reported benchmarks were captured over 500 sequential pipeline iterations to account for the inherent statistical variance of the thermodynamic key generation process. To guarantee rigorous resource profiling, each distinct cryptographic phase was executed within an isolated POSIX child process. This isolated architecture circumvents OS-level heap caching, allowing us to report accurate CPU clock times and exact high-water mark resident set size (RSS) memory allocations. Tests were executed on a system equipped with an NVIDIA RTX 3090 and an x86-64 Intel Core i9-12900K. To facilitate independent verification of these metrics, the reference implementation includes an automated, isolated benchmarking suite.

6.1 Performance Metrics

The empirical results of the 500-iteration benchmark using the `-a FAST` thermodynamic mode are detailed in Table 1. Note that the device selection (CPU vs. GPU) only impacts the key generation phase.

Table 1: QWTSS Empirical Performance Benchmarks (500 Iterations)

Phase	Hardware	Mean Time	99th Percentile	Peak RAM
Key Generation	CPU (Multicore)	2.10 s	12.17 s	0.47 MB
	GPU (CUDA)	0.43 s	1.93 s	93.56 MB*
Sign (Prover)	CPU	2.02 s	3.00 s	74.32 MB
Verify	CPU	69.8 ms	93.2 ms	20.03 MB

* RAM usage reflects the host-side CUDA context overhead. The VRAM footprint is ~0.32 MB.

Key Generation Time Variance. Unlike deterministic algorithms, thermodynamic MCMC solvers exhibit variable execution times dependent on the random walk's trajectory. Occasionally, the solver becomes deeply trapped in a local minimum, which necessitates a full thermal restart and causes the observed temporal variability. While the mean CPU key generation time is roughly 2 seconds, it exhibits a wide standard deviation typical of probabilistic solvers. However, with GPU acceleration, the mean convergence time drops to 432 milliseconds and the 99th percentile is bounded to less than 2 seconds.

[2]The complete open-source reference implementation and supporting code are available at https://github.com/axonetric/qwtss

Proving and Verification. The STARK signature generation requires approximately 2 seconds, an acceptable latency for asynchronous signing tasks. Crucially, the verification phase requires only 69.8 milliseconds. This asymmetrical computational profile—where rapid verification is decoupled from the heavier proving phase—makes QWTSS well-suited for mission-critical environments where privileged commands or software payloads must be authenticated quickly by local hardware.

6.2 Resource Utilization and Data Footprint

QWTSS exhibits a lean memory footprint compared to many conventional zk-STARK schemes. This is primarily due to the compact execution trace and the low algebraic degree of the AIR. In the reference implementation, we utilized trace flattening via intermediate columns to strictly bound the maximum algebraic degree to 4.

Table 2: QWTSS Empirical Data Footprint (500 Iterations)

Artifact	Mean Size	99th Percentile
Public Key ($\mathcal{PK}$) - JSON	0.28 KB	0.28 KB
Secret Key ($\mathcal{SK}$) - JSON	4.30 KB	4.30 KB
Signature - Raw Binary	79.68 KB	81.29 KB
Signature - JSON Base64 Envelope	106.46 KB	108.61 KB

Memory Constraints. The isolated RAM benchmarks reveal a distinct operational footprint across the cryptographic phases. On the CPU, the key generation process requires only ~0.47 MB of resident memory. This minimal footprint allows $\mathcal{SK}$ to be generated or rotated within highly memory-constrained environments, provided the associated multi-second compute latency aligns with system operational tolerances. During signature generation, the STARK prover constructs the full execution trace and FRI commitments using roughly 75 MB of peak RAM. While this substantially exceeds the runtime memory of lightweight lattice-based signatures, it is notably efficient for a zk-STARK prover.

Bandwidth. As detailed in Table 2, the underlying Wang tiling lattice allows the cryptographic keys to remain compact. The public key fits into a 287-byte JSON envelope. While the keys are minimal, the resulting raw binary STARK proof stabilizes at roughly 80 KB, corresponding to a ~136-bit STARK security level. While an 80 KB signature precludes the use of QWTSS in high-frequency or ultra-low-bandwidth network protocols, it is entirely viable for critical security transmissions where payload size is secondary to cryptographic assurance. In deployments such as high-stakes asymmetric correspondence, top-tier certificate authority (CA) root signing, or over-the-air (OTA) firmware updates for critical infrastructure, an 80 KB proof represents a manageable transmission overhead relative to the post-quantum security guarantees it provides.

7 Cryptanalysis

7.1 Taxonomy of Attack Vectors

The security of the QWTSS scheme departs from traditional algebraic hardness assumptions, relying instead on the NP-hard topological frustration of bounded quasicrystals. Consequently, standard algebraic cryptanalysis and period-finding quantum algorithms are not natively applicable. Furthermore, as demonstrated in the preceding sections, many classes of solvers are neutralized by the presence of the Overlap Gap Property, in conjunction with the critically constrained operational regime.

Therefore, the primary theoretical threat to the scheme is *structural leakage*—the potential for an adversary to extrapolate portions of the rigid quasiperiodic lattice interior from the published $\mathcal{PK}$-derived boundary conditions. We focus our cryptanalysis on a primary manifestation of this threat: the Labbé oracle boundary-projection attack (henceforth referred to as the Oracle Attack).

7.2 Oracle Attack

Aperiodic Wang tilings generated by substitution systems (such as the JR-11 alphabet) possess the Unique Composition Property (UCP), which dictates that certain zero-defect regions can be hierarchically grouped into higher-order *supertiles*—and conversely decomposed—in exactly one valid, deterministic way. They also exhibit the Local Isomorphism Property (LIP), guaranteeing that any arbitrary, zero-defect finite configuration of tiles will appear in every infinite zero-defect tiling within a predictable, finite distance.

Therefore, an adversary can execute the publicly known, deterministic Labbé algorithm to generate a sufficiently large $V \times V$ zero-defect grid (the oracle), where $V \gg 64$. By virtue of the LIP, the attacker is mathematically guaranteed to find colored edge sequences within this oracle that exactly match sub-segments of the target $\mathcal{PK}$ boundary. After locating these matches (anchors), the attacker leverages the UCP to project the adjacent deterministic tile assignments into the grid interior via spatial translation. While the QWTSS operational defect band—alongside the alien tile constraint detailed in Section 3.5—fundamentally prohibits perfect interior projections, the Labbé ground-state tiling serves as a probabilistic attractor, especially in grid regions adjacent to the boundaries.

We detail the exact geometric projection mechanics, bitmask generation, and conditional spatial filtering heuristic utilized to simulate this attack in Appendix I.

7.3 The Formalized Security Bound

The cryptographic vulnerability of this attack is formalized as the reduction in the conditional joint entropy $H(\mathcal{SK} \mid \mathcal{PK})$ of the lattice's thermodynamic state space when further constrained by the attacker's winnowed tile projections ($\mathcal{M}$). The total bits of security compromised by the Oracle Attack are defined as the differential:

$$\Delta H = H_{base} - H_{attacker} \tag{33}$$

To establish a rigorous, worst-case upper bound on ΔH, we evaluate the simulated attack under an Omniscient Threat Model. In this model, the simulated adversary is granted artificial algorithmic advantages that strictly over-approximate real-world capabilities:

1. **Omniscient Hyperparameter Sweeping:** The adversary is granted direct access to the true private key $\mathcal{SK}$ to dynamically evaluate and select the optimal minimum match length ℓ, as well as to perfectly dictate the application of the spatial filtering heuristic to maximize entropy reduction.

2. **Instantaneous Halt-State Detection:** If the attacker's candidate mask introduces unresolvable geometric contradictions, the simulation immediately detects the failure and aborts ("fast-fail"). Conversely, a real-world adversary could expend considerable computational resources exploring a mathematically impossible state subspace before triggering an arbitrary halting heuristic.

3. **Defect Density Omniscience:** The simulated attacker is granted exact knowledge of the $\mathcal{SK}$ defect count. This permits perfect calibration of the stochastic mask relaxation rate and optimal temperature targeting for the AIS schedule.

Consequently, the empirical joint entropy reductions reported in our analysis represent a strict, theoretical ceiling on an adversary's capabilities with this attack vector.

7.4 Empirical Results and Security Margins

The empirical application of the simulated Oracle Attack reveals a substantial entropy reduction against the target $\mathcal{SK}$. As illustrated in Figure 15, the Labbé anchor-and-project methodology successfully compromises a portion of the configuration space, reducing the joint entropy (ΔH) by $\approx$ 340 bits at QWTSS operational parameters $D = 160$ and $\rho_B = 0.70$. The majority of this algorithmically pruned entropy is due to strong tiling correlations directly adjacent to the boundaries; the central interior of the lattice remains largely immune to the oracle's deterministic projections. Geometrically, this entropy reduction is equivalent to an attacker effectively shrinking the original 64×64 cryptographic grid down to an uncompromised 51×51 grid core (maintaining the original, uniform entropy density).

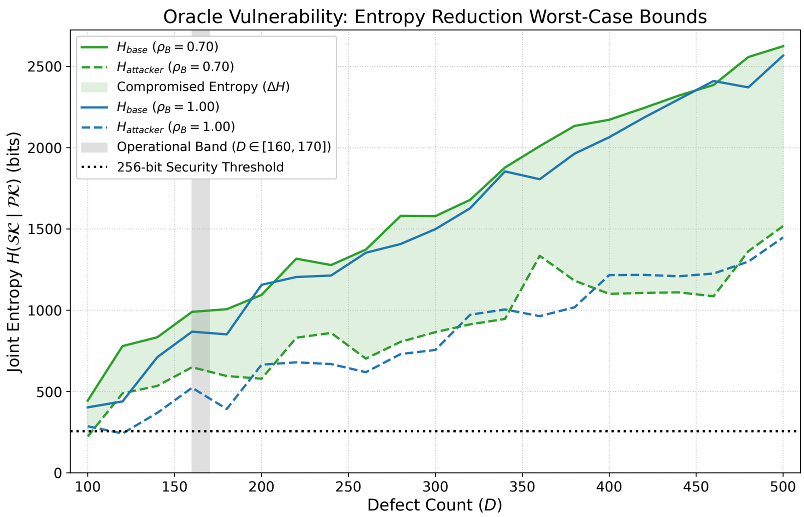

Figure 15: Empirical bounds of the Oracle Attack demonstrating the reduction in $\mathcal{PK}$-conditioned joint entropy. Solid lines denote the baseline cryptographic security (H_{base}), while dashed lines represent the maximally constrained entropy ($H_{attacker}$) achieved by the simulated adversary. The shaded green region illustrates the total bits of security compromised (ΔH) under the default boundary pinning density ($\rho_B = 0.70$). Within the QWTSS operational band (shaded gray), the remaining joint entropy comfortably exceeds the 256-bit post-quantum security threshold (black dotted line) under the worst-case Omniscient Threat Model.

Crucially, within the scheme's designated operational band ($D \in [160, 170]$), the adversary's maximally constrained joint entropy ($H_{attacker}$) remains at $\approx$ 600 bits. This surviving cryptographic margin remains well above the standard 256-bit threshold targeting post-quantum symmetric security. The empirical $H_{attacker}$ trajectory exhibits stable, linear scaling, indicating predictable bounding behavior. Additionally, we observe that the $\rho_B = 0.70$ metric maintains a positive entropic offset relative to $\rho_B = 1.00$, further justifying its role in the scheme.

Ultimately, the system's security rests on the intractable combinatorics of the grid's internal defect distribution, securely sampled through the joint entropy minimization of the annealing process. These defects are mathematically opaque to deterministic boundary projections, ensuring that the surviving lattice core constitutes a computationally intractable search space.

8 Conclusion

8.1 Summary of Cryptographic Security

The post-quantum security of the QWTSS architecture relies on a synthesis of statistical mechanics and zero-knowledge arithmetization. The thermodynamic domain resides deep within the frustrated glassy phase. Aperiodic Wang tilings bounded by specific perimeter constraints exhibit no known cyclic structure, period-finding geometry, or hidden subgroups. Consequently, Shor's algorithm and its derivatives are inapplicable.

Fundamentally, QWTSS derives its security from a composite barrier: the preimage resistance of a 504-bit Poseidon sponge hash gated behind an NP-hard combinatorial search. The glassy state does not act as a standalone security mechanism—since an attacker can leverage the same annealing process—but rather functions as an asymmetric work-factor multiplier. QWTSS forces the adversary to navigate a frustrated thermodynamic landscape to generate each candidate grid. Instead of standard hash-based security, every single bit of the Poseidon digest is uniformly anchored in the underlying NP-hard problem space.

The annealing process natively enforces a baseline computational work requirement at least 6 decimal orders of magnitude greater than the hash primitive itself. Accordingly, the total classical QWTSS cryptographic resistance is defined by the product of the preimage search space and the forced operational interaction with the problem domain (i.e., the work factor $\mathcal{W}$):

$$\mathcal{W}_{\text{attack}} \approx O(2^{504} \cdot \mathcal{W}_{\text{anneal}}) \tag{34}$$

where $\mathcal{W}_{\text{anneal}}$ represents the discrete computational effort (measured in thermodynamic state transitions or equivalent gate operations) required to resolve one valid grid.

Even if a future quantum adversary applies Grover's algorithm to reduce the effective preimage security to 2^{252} operations, the simulated thermodynamic work enforced by $\mathcal{W}_{\text{anneal}}$ incurs a mandatory computational penalty. By anchoring a 504-bit cryptographic hash within a thermodynamic domain conservatively bounded at ≈ 600 bits of classical security against known cryptanalytic techniques, QWTSS achieves a robust baseline of post-quantum security.

8.2 Potential Applications

The primary operational constraints of the QWTSS scheme are: (1) the highly variable, multi-second key generation latency, and (2) the relatively large 80 KB signature size. Because of these properties, QWTSS is not suitable for validating high-frequency everyday internet traffic or driving footprint-conscious Layer-1 blockchain protocols.

However, as the threat landscape evolves with Cryptographically Relevant Quantum Computers (CRQCs) and AI-assisted cryptanalysis, structurally diverse and resilient digital signatures will become increasingly vital. Historical precedent suggests that unforeseen analytical breaks may occur in schemes currently believed to be secure. In a landscape where the threat model includes state-sponsored adversaries equipped with quantum resources, securing critical military, governmental, and corporate communications against impersonation warrants architectures built on composite hardness assumptions with conservative security margins.

For these environments, where signature payload size and key generation latency are secondary to the requirement of post-quantum authenticity, QWTSS offers a robust alternative. Practical QWTSS deployments include:

- **Mission-Critical Infrastructure:** Secure boot sequences and root-of-trust operations within air-gapped hardware enclaves or embedded cryptographic modules, where key generation and rotation are infrequent relative to continuous, rapid verification.
- **High-Stakes Communications:** Military command-and-control (C2) systems, sovereign and corporate financial authorizations, and asynchronous secure correspondence, where a multi-second signing delay is an acceptable trade-off for robust forgery resistance.

- **Broadcast Authentication:** Over-the-air (OTA) firmware updates for critical infrastructure (e.g., power grids, orbital satellites, or autonomous vehicles), where an 80 KB proof represents a minimal overhead relative to megabyte-scale payloads.
- **Batched Verification:** Layer-2 rollup architectures, where a single 80 KB STARK proof can authenticate the computational integrity of thousands of batched state transitions, amortizing the bandwidth cost across large aggregate data sets.

By decoupling a mathematically bounded, rapid verification phase from a deliberately constrained, thermodynamically hard generation phase, QWTSS provides a viable alternative for robust post-quantum digital signatures.

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A Poseidon Sponge Hash Security

The QWTSS scheme leverages the Poseidon hash function due to its high efficiency in ZK proofs, as it is optimized for large prime fields. Specifically, we instantiate a standard Poseidon-1 sponge construction over the 252-bit STARK prime field ($\mathbb{F}_p$ where $p = 2^{251} + 17 \times 2^{192} + 1$), chosen for native compatibility with the Stone Prover. To achieve a 504-bit security digest, the permutation operates on a state width of $t = 4$ (two capacity elements and two rate elements), utilizing a standard 4×4 Maximum Distance Separable (MDS) matrix for optimal diffusion.

Because arithmetization-oriented hash functions rely on low-degree S-boxes (specifically x^3 in our implementation) to minimize AIR constraint degrees, they are subject to scrutiny regarding algebraic cryptanalysis, such as Gröbner basis computation and interpolation attacks. In isolation, a low-degree permutation requires a carefully calibrated number of full (R_F) and partial (R_P) rounds to ensure the algebraic degree of the system reaches the maximum possible threshold.

However, the specific context in which Poseidon is deployed in our scheme provides a robust, structural defense. Rather than hashing a short, fixed-length preimage, the function acts as an iterated sequential accumulator over the $L \times L$ grid. For a 64×64 grid, the sponge processes a continuous chain of 4,096 steps, where the specific tile ID and the current spatial step counter are injected into the two rate elements, contributing unique algebraic constraints and strict domain separation at each step. Crucially, because the verification is zero-knowledge, none of the intermediate 4,095 sponge states are ever revealed. An adversary attempting an algebraic attack must therefore model the continuous composition of the hash state across all 4,096 grid positions. Because polynomial degrees multiply across composed operations, executing a sequence of 4,096 consecutive Poseidon rounds yields a composite system with an astronomically high algebraic degree.

For a Poseidon sponge utilizing an S-box of degree $d = 3$ over our 4,096-step grid verification, the theoretical degree evaluates to:

$$\deg(S_{4096}) = 3^{4096} = (2^{\log_2 3})^{4096} \approx 2^{6,492} \tag{35}$$

Because the accumulator operates over $\mathbb{F}_p$, the maximum possible algebraic degree of any polynomial is bounded by the field order ($p - 1 \approx 2^{252}$).

$$2^{6,492} \gg |\mathbb{F}_p| \approx 2^{252} \tag{36}$$

The accumulator reaches algebraic saturation (exceeding a degree of 2^{252}) after only $R = 159$ rounds. By enforcing a continuous chain of 4,096 full rounds, QWTSS reaches maximum field complexity 25 times over. This rapid degree saturation provides strong theoretical assurances that the transition function resists algebraic attacks, rendering Gröbner basis algorithms, interpolation attacks, and higher-order differential cryptanalysis strictly more expensive than a brute-force search.

While the leading edge and interior of the grid are mathematically protected by this continuous degree compounding, the AIR enforces 64 explicit blanking rounds at the end of the trace to cryptographically seal the trailing edge of the hash. By forcing the sponge to absorb a zero value while the spatial constraints are deactivated, the final surviving lattice variables are pushed through 64 unmalleable, deterministic S-boxes. This compounds the trailing algebraic degree by an additional $3^{64} \approx 3.4 \times 10^{30}$. This dense, highly non-linear mathematical wall guarantees that the entire hash chain, including the very last malleable elements, is secure against any type of algebraic forgery attack.

B Annealed Importance Sampling (AIS)

We define the partition function Z of the Wang tiling grid over the alphabet $\mathcal{A}$ using standard thermodynamic notation:

$$Z = \sum_{\sigma \in \Omega} e^{-\beta E(\sigma)} \tag{37}$$

AIS provides a mathematically rigorous stochastic Monte Carlo estimator for the ratio of partition functions between a tractable, unconstrained base distribution and the highly constrained target distribution. Specifically, AIS functions as an algorithmic realization of Jarzynski's Equality [16], which relates the non-equilibrium work performed during a finite-time phase transition to the equilibrium free energy difference between states.

By defining a sequence of K intermediate distributions, AIS smoothly bridges the state space. Let $f_0(\sigma)$ be the unnormalized density of an easily sampled initial state (e.g., a high-temperature independent tile distribution with a known partition function Z_0), and let $f_K(\sigma)$ be the highly constrained target cryptographic distribution ($Z_K = Z$). The intermediate distributions are defined by an inverse temperature annealing schedule, $0 = \beta_0 < \beta_1 < \cdots < \beta_K = 1$, such that:

$$f_k(\sigma) = f_0(\sigma)^{1-\beta_k} f_K(\sigma)^{\beta_k} \tag{38}$$

For each independent AIS chain i, a sample σ is generated by transitioning through the sequence of distributions using MCMC kernels $T_k(\sigma' \mid \sigma)$ that leave f_k invariant. The crucial metric extracted from this process is the importance weight, $w^{(i)}$, accumulated along the trajectory, acting as the Jarzynski work term:

$$w^{(i)} = \prod_{k=1}^{K} \frac{f_k(\sigma_{k-1}^{(i)})}{f_{k-1}(\sigma_{k-1}^{(i)})} \tag{39}$$

Since Jarzynski's Equality and AIS guarantee that the expected value of these non-equilibrium weights converges to the ratio of the equilibrium partition functions, we can approximate the target partition function Z_K via an ensemble of N independent chains:

$$\frac{Z_K}{Z_0} \approx \frac{1}{N} \sum_{i=1}^{N} w^{(i)} \tag{40}$$

Because the target distribution uniformly weights all valid configurations that satisfy the QWTSS public key and defect constraints, the total number of valid states is exactly Z_K. Thus, the joint entropy, H, of the cryptographic state space is directly derived by taking the base-2 logarithm of the estimated partition function:

$$H = \log_2(Z_K) \approx \log_2\left(Z_0 \cdot \frac{1}{N} \sum_{i=1}^{N} w^{(i)}\right) \tag{41}$$

C Extreme Value Theory (EVT) Asymptotic Estimation

Under Extreme Value Theory, the extreme minimums of a distribution bounded from below must strictly converge to a Type III (Weibull) extreme value distribution according to the Fisher-Tippett-Gnedenko theorem. To formally model the expected global minimum Hamming distance $HD_{min}(N)$ as a power-law decaying toward an absolute mathematical asymptote c, we use the following equation:

$$\mathbb{E}[HD_{min}(N)] \approx c + A \cdot N^{-b} \tag{42}$$

where c represents the asymptotic floor of the Overlap Gap (Δ_{OGP}), A is the scale parameter, and $b = 1/k$ is derived from the Weibull shape parameter k of the extreme left tail.

To calculate the asymptote c estimates for the three deepest search runs ($N \approx 3.0 \times 10^{10}$ comparisons), we isolated the empirical drop sequences and applied a Non-linear Least Squares (NLS) optimizer utilizing a bounded grid-search. While the scale and shape parameters for each candidate were efficiently derived via log-log linearization, the algorithm iteratively searched for the final asymptote c that minimized the weighted Residual Sum of Squares (RSS) of the untransformed residuals in linear space.

D Energy Barrier Analysis and SFPGS

To analyze the transitional state space between two *relatively* nearby $\mathcal{PK}$-conditioned private keys ($\mathcal{SK}$), we performed the following experiment:

1. Generate an ensemble of 1000 valid, independent $\mathcal{PK}$-conditioned $\mathcal{SK}$s. Calculate the pairwise Hamming distances between all candidates to find the global minimum distance, HD_{min}. We define this closest pair as $(\mathcal{SK}_A, \mathcal{SK}_B)$.

2. Execute a specialized Stochastic Flat-Path Greedy Solver (SFPGS) to determine a near-optimal, 1-step global lookahead path to transition $\mathcal{SK}_A \rightarrow \mathcal{SK}_B$.

3. Repeat Step 2 a total of 1000 times to generate an ensemble of unique, near-optimal transition trajectories (stochastic greedy walks) through the phase space.

Importantly, the SFPGS is gifted with two artificial omniscience advantages designed to find the path of least resistance:

- **Target Omniscience (Routing):** The solver is aware of the exact per-tile configuration of the destination $\mathcal{SK}_B$. At each step, it is restricted to only flipping a grid tile that does not currently match $\mathcal{SK}_B$. This guarantees monotonic progress toward the destination and strict convergence in exactly HD_{min} steps.

- **Selection Omniscience (Greedy Neighborhood Lookahead):** The solver evaluates all remaining eligible tile flips at each step. Rather than descending the energy gradient, it ranks moves by their absolute deviation from the starting energy state (D_{start}), heavily favoring moves that maintain a "flat" trajectory (i.e., minimizing $|\Delta E|$). This allows the solver to cherry-pick the path of least thermodynamic resistance.

The rationale for the flat-path heuristic is that minimizing D (defect count) early in the pathway only increases the topological rigidity of the grid, empirically ensuring an even steeper energy barrier (climb "out of the valley") during the remaining stages of the transition. Similar to how roads in mountainous regions follow contour lines, the SFPGS attempts to navigate isentropic manifolds to discover any continuous, easily traversable corridors between the two keys. To more robustly explore the space around these possible corridors, the SFPGS employs a Top-5 stochastic sampling rule. At each step, it randomly selects its next move uniformly from the 5 highest-ranked flat-path options.

1-to-1 Stochastic Transition Visualization. To capture the stochastic variance of the SFPGS transitions, we first map the specific phase space corridor between a single pair of keys ($\mathcal{SK}_A \to \mathcal{SK}_B$) using a 2D path density histogram (Figure 8). The x-axis represents the normalized transition progress, while the y-axis tracks the system energy (D). Because the solver employs a Top-5 stochastic sampling rule over 1000 independent walks, color intensity is plotted on a logarithmic scale to highlight faint, low-probability deviations against the denser core of standard trajectories. Visually, the transition begins as a bright, narrow path where SFPGS is able to navigate within isentropic space. However, as the solver exhausts the available flat-path permutations (typically starting at $t_{norm} \gtrapprox 0.2$), it is inevitably forced into a rugged energy landscape represented by the diffuse band in higher D regions (the energy barrier). The solver is forced to scramble over varying topological dead-ends to reach $\mathcal{SK}_B$. This logarithmic heatmap provides strong visual evidence that valid, near-optimal solution clusters do not form a continuous, easily traversable manifold.

1-to-N Deterministic Transition Visualization. To expand this analysis from single 1-to-1 corridors to a global state space topology, we execute a 1-to-N radial mapping (Figure 9). In this visualization, a single master key ($\mathcal{SK}_{master}$) is locked at the origin ($r = 0$) of a polar coordinate system, while 100 valid, independent candidate keys (boundary-conditioned on $\mathcal{SK}_{master}$) are distributed along the circumference ($r = 1.0$) radially in random order. The SFPGS executes independent flat-path walks radiating outward from the center to each target, using a deterministic Top-1 selection rule to choose the single best greedy path possible. The resulting 2D topographical energy map clamps the lowest color gradient to $D = 160$ and applies a 90th-percentile clipping threshold to retain intermediate-level gradient detail. This radial view reinforces that continuous, easily traversable manifolds between various $\mathcal{PK}$-conditioned $\mathcal{SK}$ do not exist. Instead, we again find robust evidence of the OGP within the QWTSS operational band.

Activation Energy Barrier. To establish a lower bound of the thermodynamic obstacle impeding the transition between two *relatively* "nearby" solution clusters, we collect statistics over an ensemble of SFPGS-generated trajectories following the deterministic Top-1 selection rule, using a fresh independent ($\mathcal{SK}_A, \mathcal{SK}_B$) closest pair for each generated path. We formally define the *Activation Energy Barrier* (ΔE_{act}). Let γ represent a specific transition trajectory consisting of a discrete sequence of valid tile flips, $\gamma = (\mathcal{SK}_0, \mathcal{SK}_1, \ldots, \mathcal{SK}_L)$, transitioning from the starting grid to the target. If $D(\mathcal{SK}_i)$ represents the system energy (defect count) at step i, the activation barrier of the path is defined as the absolute difference between the highest thermodynamic peak and the lowest thermodynamic valley encountered along the trajectory:

$$\Delta E_{act}(\gamma) = \max_{0 \le i \le L} D(\mathcal{SK}_i) - \min_{0 \le i \le L} D(\mathcal{SK}_i) \tag{43}$$

This metric isolates the true energy barrier of the Overlap Gap, representing the peak thermodynamic excursion required to successfully cross the "dead zone" separating independent solution clusters. We observe the absolute minimum empirical excursion to be $\Delta E^*_{act}\big|_{\text{SFPGS}} \approx 20$, and an average value at $\Delta E_{act}\big|_{\text{SFPGS}} \approx 42.7$. In the context of solvers without the omniscience advantages of SFPGS, the practical value is much higher. We conclude that the Overlap Gap region functions as an impenetrable thermodynamic barrier.

E Rosenbluth-Knuth Estimation Method with Ray-Casting

To prevent path-looping and ensure valid structural adherence, each stochastic ray (probe) is governed by two strict topological constraints:

1. **Strict Monotonicity:** The ray must exclusively select tile permutations that monotonically increase the Hamming distance from the origin anchor at each step.

2. **Defect Confinement:** Every step must strictly maintain the overall grid energy (the defect count) within the operational band ($160 \le D \le 170$).

Furthermore, because the Wang tilings allow for spatially independent permutations, multiple distinct pathways inevitably converge on identical grid states. Due to the severely suppressed average branching factor of the JR-11 grammar (25), the standard thermodynamic correction for this redundancy (dividing by the step factorial $N!$) would catastrophically under-estimate the true entropy. To accurately resolve this commutativity without the computationally intractable burden of global graph tracking, we treat the phase space as a layered directed acyclic graph. We calculate the exact topological weight W of each terminal ray using a modified Rosenbluth estimator applied to the self-avoiding walk:

$$W = \prod_{i=0}^{N-1} \frac{d_{out}(x_i)}{d_{in}(x_{i+1})} \tag{44}$$

where $d_{out}(x_i)$ represents the available forward combinatorial capacity (out-degree) at step i, and $d_{in}(x_{i+1})$ represents the exact local commutativity (in-degree) of the subsequent state. We measure d_{in} dynamically at each step during the walk via an empirical 1-step topological look-back probe. Crucially, auxiliary deep recursive profiling of the cluster manifolds confirmed strict backward-convexity—demonstrating that locally valid parent states exhibit a 100% survival rate back to the origin, at every step and for every ray. This finding ensures the 1-step d_{in} penalty is entirely free of topological pruning bias, allowing us to accurately isolate the true net geometric entropy of the ray's path. The cluster's total entropic volume is then bounded via a LogSumExp aggregation of these dynamically corrected terminal ray weights.

Anchor Centrality Bias. While the volume estimate for any single anchor converges reliably under high ray counts, the magnitude of this estimate is acutely sensitive to the anchor's specific location within the highly anisotropic cluster manifold. Because the stochastic ray-casting is strictly monotonic, it suffers from a local topological bias: anchors situated deep within peripheral dendrites yield severely truncated volume estimates. Here, outward rays are overwhelmingly trapped within local, fractal dead-end branches before they can ever reach the larger cluster interior. Conversely, anchors near the cluster's massive principal axis afford their rays maximal topological reach. Therefore, to capture the true thermodynamic volume, we employ a long-range, non-monotonic intra-cluster MCMC walk to gather an ensemble of N geometrically decorrelated anchors. With sufficient diffusion distance, the ensemble elements naturally gravitate toward the cluster's maximally entropic core. The absolute lower-bound entropic volume of the target cluster is then established as the maximum estimate acquired across this N-anchor ensemble.

F Total Number of Distinct Solution Clusters

To derive the estimated total number of distinct solution clusters $N_{clusters}$, we must integrate the cluster frequency distribution over the empirical phase space. As established, the combinatorial mass of the phase space is governed by a log-normal distribution. Let $P(H)$ be the probability density function of the sampled logarithmic volumes $H \sim \mathcal{N}(\mu, \sigma^2)$. Because our random anchor sampling is proportional to cluster volume, $P(H)$ represents the fraction of the total solution space volume V_{total} occupied by all clusters of scale H:

$$V(H) = V_{total} \cdot P(H) \tag{45}$$

Because a single cluster at scale H contains exactly 2^H states, the raw frequency distribution of the clusters $N(H)$ is this scale-specific volume divided by the cluster size:

$$N(H) = \frac{V(H)}{2^H} = V_{total} \cdot P(H) \cdot 2^{-H} \tag{46}$$

The total number of distinct clusters in the global phase space is the integral of this frequency distribution from the empirical minimum boundary H_{min} to infinity. We integrate to the thermodynamic limit because the exponential penalty of 2^{-H} renders cluster frequency at the extreme right tail functionally zero:

$$N_{clusters} = V_{total} \int_{H_{min}}^{\infty} P(H) 2^{-H} \, dH \tag{47}$$

Substituting the normal probability density $P(H) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(H-\mu)^2}{2\sigma^2}}$, we evaluate the integrand's exponent:

$$-\frac{(H-\mu)^2}{2\sigma^2} - H \ln 2 \tag{48}$$

By completing the square, we can factor out the non-integrable constants:

$$-\frac{(H-\mu')^2}{2\sigma^2} - \mu \ln 2 + \frac{\sigma^2 \ln^2 2}{2} \tag{49}$$

where $\mu' = \mu - \sigma^2 \ln 2$, yielding a shifted Gaussian integral:

$$N_{clusters} = V_{total} \, 2^{-\mu + \frac{\sigma^2 \ln 2}{2}} \int_{H_{min}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(H-\mu')^2}{2\sigma^2}} \, dH \tag{50}$$

This integral represents the complementary cumulative distribution function (the right tail) of the shifted Gaussian. Because our empirical floor H_{min} is significantly larger than the shifted mean μ', we can apply the standard asymptotic tail expansion $1 - \Phi(z) \approx \frac{1}{z\sqrt{2\pi}} e^{-z^2/2}$ (where $z = \frac{H_{min}-\mu'}{\sigma}$). Strikingly, rearranging the expanded terms collapses the equation mathematically back into the original probability density function evaluated at the boundary:

$$N_{clusters} \approx V_{total} \cdot 2^{-H_{min}} \cdot P(H_{min}) \cdot W \tag{51}$$

where W is the effective entropic integration width:

$$W = \frac{\sigma^2}{H_{min} - \mu + \sigma^2 \ln 2} \tag{52}$$

This analytical reduction proves that the sheer multiplicity of the state space is overwhelmingly dominated by the absolute smallest clusters at the H_{min} boundary, weighted by their entropic width W.

We apply our empirical phase space parameters: $V_{total} = 2^{940}$, $\mu = 137.5$ bits, $\sigma = 27.2$ bits, and the empirically observed lower bound $H_{min} = 84$ bits. Evaluating the effective width yields $W \approx 1.61 \approx 2^{0.69}$ bits. The density of states at the empirical floor is $P(84) \approx 0.0021 \approx 2^{-8.88}$.

Substituting these values yields the estimated total count of uniquely isolated QWTSS solution clusters:

$$N_{clusters} \approx 2^{940} \times 2^{-84} \times 2^{-8.88} \times 2^{0.69} \approx 2^{847.81} \tag{53}$$

G Algorithmic Optimizations for Key Generation

To achieve practical, computationally efficient, sub-second key generation, we implemented an optimized, GPU-accelerated MCMC solver. While the core spatial update mechanism preserves the instantaneous detailed balance of the Markov chain, the initialization and temporal cooling heuristics are explicitly tuned for rapid convergence. All empirical data presented in this paper were generated with these optimizations enabled. We highlight five essential algorithmic details here:

1. Parallel Red-Black Heat Bath. We observed that a parallel Red-Black checkerboard update scheme utilizing Heat Bath (Gibbs) sampling clearly outperformed standard one-by-one Metropolis-Hastings mutations. Because the local energy of any given tile depends strictly on its immediate von Neumann neighborhood, tiles sharing the same checkerboard parity are conditionally independent at any single update step. This topological property allows thousands of GPU threads to evaluate the full 11-tile probability distribution and independently sample new states across the grid simultaneously. Crucially, alternating between red and black phases guarantees that no two adjacent tiles mutate concurrently. While this composite update breaks strict detailed balance within the Markov chain, it preserves global balance, which is sufficient to guarantee proper convergence to thermodynamic equilibrium. This optimization is applicable to both GPU parallel processing and CPU multithreading. When used in conjunction with CUDA's `__constant__` memory primitive, we realized significant reductions in key generation latency on the GPU.

2. Isotropic Greedy Pre-Quench. Rather than executing a purely geometric cooling schedule from infinite temperature, the solver begins with an isotropic greedy pre-quench phase. The grid is subjected to a brief, near-zero temperature ($\beta = 20.0$), forcing all parallel threads to act as purely greedy agents. This rapidly collapses the grid from a highly chaotic liquid state into the initial stages of the glassy phase, establishing multiple distributed nucleation sites. This noticeably reduces the high-energy thermalization work required, and empirically improves the spatial defect distribution to be slightly more uniform. This optimization can be deactivated via the `./qwtss-cli keygen -a ADIABATIC` switch in the reference executable.

3. Annealing Hyperparameter Tuning. The average computational effort required to converge on a valid $\mathcal{SK}$ also depends on the values of certain annealing hyperparameters. These values are often interdependent, and are also dependent on QWTSS metaparameters such as $L = 64$, $160 \leq D \leq 170$, etc. While we do not claim these values represent the globally optimal set, they are the result of our empirical tuning:

1. `cooling_rate` = 0.99994 — The annealing process is extremely sensitive to this parameter. Values outside a very thin working band will cause complete convergence failure.
2. `reheat_temp` = 0.39 — Upon detection of convergence stagnation (topological locking), it is optimal to reheat just above the glassy phase transition to avoid extra work.
3. `check_interval` = 2200 — This value dictates how many update steps occur between each defect counting and convergence checking event. In a parallel computing context,

these $O(N^2)$ events are expensive (for the GPU, PCIe bus latency for host-side syncs; for the CPU, atomic thread contention). Both over-checking and under-checking can degrade performance. We found the optimal value is highly sensitive to the target D band.

4. `stagnation_patience` = 2 — This value controls how many consecutive check events elapse before convergence stagnation (a failure to reach a lower D level than the previous check) is triggered and the `reheat_temp` is applied.

5. `max_steps` = $250,000$ — We abort the annealing process completely after `max_steps`. Empirical data shows that executing a Las Vegas Restart Strategy is optimal after this threshold, rather than attempting additional reheating cycles.

4. Cryptographic PRNG Architecture. To ensure the security of the MCMC thermodynamic trajectory, the key generation pipeline utilizes a hybrid pseudo-random number generator (PRNG) architecture. On the host CPU, the ChaCha20 stream cipher is used to securely expand the 256-bit hash seed, avoiding the state-space bottlenecks and linear predictability of standard generators like the Mersenne Twister. However, implementing a 512-bit state CSPRNG like ChaCha20 directly within the highly parallel GPU heat bath kernel induces severe register pressure, thus crippling thread occupancy and throughput. Instead, the device leverages `Philox4x32-10`, a counter-based PRNG native to cuRAND that offers statistically flawless uniformity (passing TestU01 BigCrush) with minimal register overhead. To secure the device execution, the host-side ChaCha20 instance performs comprehensive entropy injection, generating a unique, cryptographically secure 64-bit seed for every individual GPU thread. This injects $262,144$ bits of initial entropy into the 64×64 grid, mitigating brute-force attacks on the annealing trajectory while preserving maximum hardware performance.

5. Fixed-Point MCMC Mathematics. In standard simulated annealing, evaluating the Boltzmann distribution requires expensive floating-point exponentiation. However, in the QWTSS grid topology, the local energy differential ΔE for any single tile mutation is strictly bounded to a discrete integer set ($0 \leq \Delta E \leq 4$). Exploiting this finite domain, the solver extracts all floating-point overhead from the highly parallel worker threads. Instead, the host dynamically pre-computes a small lookup table (LUT) of exact Boltzmann weights for the current temperature, scaled into a fixed-point integer format. The GPU and CPU heat-bath kernels then sample the thermodynamic distribution using exclusively branchless integer arithmetic and fast bitshift range reductions. Beyond bypassing `expf()` calls—which yields disproportionately large latency improvements on the CPU compared to the GPU's natively optimized FP32 pipelines—this pure-integer architecture guarantees 100% cross-platform deterministic key generation by completely eliminating hardware-specific floating-point rounding discrepancies.

H Work Circumvention Vectors

Here we consider deterministic methods that an impatient prover could utilize to bypass the simulated thermodynamic work required for secure $\mathcal{SK}$ generation. The QWTSS operational defect range ($D \in [160, 170]$) contains sufficient degrees of freedom to synthetically decouple the boundary constraints from the grid's interior. The interior can then be easily "painted" with a zero-defect tiling using Labbé's Wang tile formalization. After the dishonest prover walls off the problematic border segments with manually placed defects (e.g., resembling a picture frame), any additional defects needed to meet the operational band requirements can be scattered uniformly via single-tile mutations, which can be carefully selected to provide as few as two additional defects per flip.

We simulated synthetically generated grids using a wide variety of border segment decoupling strategies. These topologies included straight walls, jagged walls, diagonals/staircases, inner squares, inner diamonds, radial partitions, pure plane flooding, and Voronoi partitioning schemes utilizing various distance metrics (L_1, L_2, and Chebyshev). To model a rational adversary performing gradient descent across this complex penalty space, our profiling employed multiple greedy 1-step lookahead optimizers, testing both Total Constraint Violation (TCV) and weighted action scoring frameworks. These optimizers were provided with perfect knowledge of all constraints and their exact threshold bounds during each simulated attempt.

Through iterative Leave-One-Out (LOO) constraint analysis over hundreds of thousands of sampled heuristic attempts, we discovered that the three synergistic constraints outlined in Section 3.5 perfectly discriminate between naturally annealed grids and artificially cobbled deterministic grids that circumvent the simulated thermodynamic work:

1. **Macroscopic Divergence** (Alien_Tiles $\geq$ 1500): A requirement that a large portion of the grid's tiles must diverge from the $\mathcal{PK}$-derived oracle grids. This ensures the dishonest prover must construct a large anomalous area, consuming the majority of their defect budget just to address its perimeter.

2. **Core Entropy** (Core32 $\geq$ 20): A requirement that a minimum number of defects must manifest in the central interior of the grid (32×32 core region). Deterministic algorithms naturally push edge mismatches outward near the boundaries; this constraint forces the dishonest prover to also expend their limited defect budget in the center.

3. **Orthogonal Smoothness** (Line_Max $\leq$ 12): A strict limit on the maximum number of defects that can share any single horizontal or vertical seam. Instead of using optimal straight lines to wall off areas, the dishonest prover is forced to use staircases or diagonals, which again consumes valuable defects (especially in 2D Manhattan distance).

The absolute security of this defense relies not on any single constraint, but on the mutually exclusive geometric paradox they create when enforced simultaneously, alongside the strict upper bound on the total allowable defect budget ($D \leq 170$). To satisfy the alien tiles count, the dishonest prover may introduce a phase offset to the zero-defect tiling; however, this introduces numerous new defects along the sheared edges, consuming a large portion of the defect budget. To satisfy the core entropy requirement, they must inject isolated defects into the center of the grid, further depleting the budget. Finally, the orthogonal smoothness limit prevents defects from being efficiently aligned in the same vertical or horizontal seam.

These three constraints implement a synergistic geometric intersection that isolates two characteristic properties that only appear together in naturally annealed grids (Figure 16): (1) fairly uniform defect distributions, both orthogonally and globally, and (2) significant macroscopic divergence from the deterministic zero-defect tilings. To satisfy these constraints deterministically, the dishonest prover requires more degrees of freedom than the operational band grants. However, a thermodynamic annealer satisfies these constraints naturally via global entropy minimization. Importantly, these constraints enforced jointly allow an empirical acceptance rate of $\approx 95\%$ on legitimately annealed keys, producing little friction for honest provers.

We empirically verified the safety margins of these constraints against our suite of synthetically optimized topologies. The most critical limit is the upper bound on the total defect count D. Through exhaustive search, we found these synergistic constraints first fail at $D \approx 190$ allowed defects, where $5/282,000$ sampled circumvention attempts successfully bypassed the constraint set. Finally, we note that these synergistic constraints need not be failsafe in the absolute sense (as forgery resistance is already established without them). They simply need to enforce a greater computational cost than a single annealing run requires.

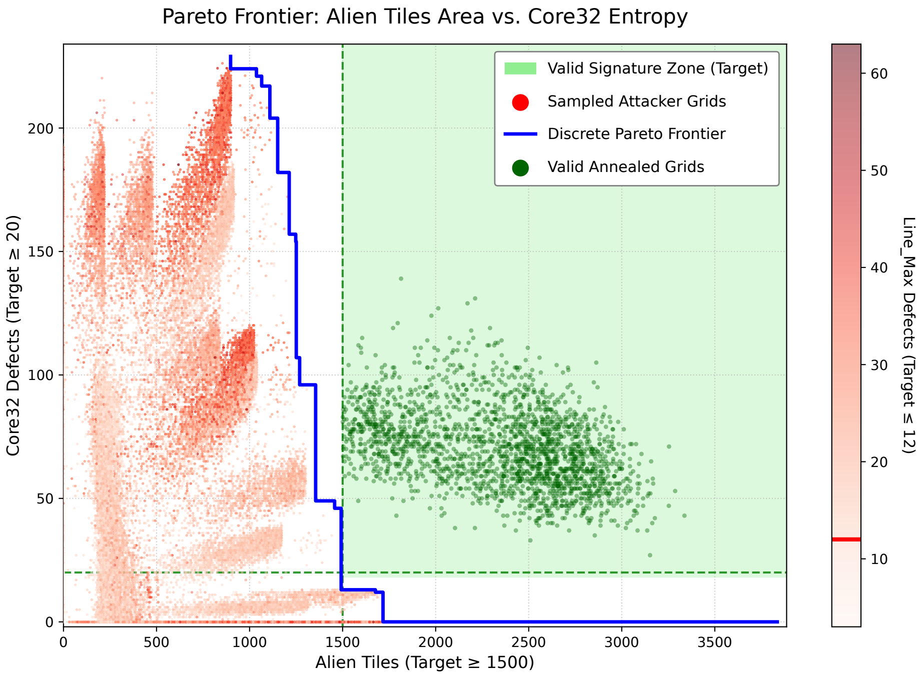

Figure 16: The discrete Pareto frontier across a 2D slice of the anti-circumvention constraints (Alien Tiles vs. Core32 Defects). The valid zone (green) illustrates the mutually exclusive nature of these synergistic constraints applied to a dishonest prover's deterministically optimized grids.

I Oracle Attack Simulation

To quantify the Oracle Attack vulnerability bounds defined in Section 7.2, we modeled the adversary's anchor-and-project mechanism and empirically calculated the $\mathcal{PK}$-conditioned joint entropy differential (ΔH).

Oracle Anchoring and Geometric Projection. The simulated attack begins by treating the $\mathcal{PK}$ boundary edge colors as four independent strings of length L. The attacker searches the pre-computed $V \times V$ oracle grid to find all occurrences of these strings of at least a minimum match length ℓ. Each match is formally defined as an anchor $\alpha = (x, y, e, o)$, where (x, y) denotes the spatial coordinates within the oracle, $e \in \{N, S, E, W\}$ specifies the boundary edge, and o represents the linear offset of the substring match.

Because small values of ℓ yield a combinatorial explosion of spurious matches, we restrict the simulated adversary to a maximum retention of $500,000$ anchors per boundary edge. Empirical profiling reveals this limit is highly conservative; peak attack efficacy is consistently achieved at intermediate ℓ values (typically $20 \le \ell \le 40$), where the anchor volume naturally falls well below this threshold.

For each anchor found, the attacker projects the corresponding 64×64 interior from the oracle grid into a master candidate bitmask $\mathcal{M}$. Because the alphabet size $|\mathcal{A}|$ is small ($|\mathcal{A}| \le 16$), $\mathcal{M}$ is efficiently represented as a grid of 16-bit integers. For a given spatial index $i \in \{0, \dots, 64^2 - 1\}$, the t-th bit of $\mathcal{M}_i$ is set to 1 if tile ID t is predicted by any anchor covering that coordinate. Consequently, for a grid evaluated against k overlapping anchors, the final candidate mask at index i is the bitwise OR of all individual anchor projections:

$$\mathcal{M}_i = \bigvee_{j=1}^{k} \mathrm{Proj}(\alpha_j)_i \tag{54}$$

Spatial Filtering and Corner Pinning. To evaluate the maximum possible precision of the attack, we implemented an optional 2D spatial filter. This heuristic assumes that a valid private key is most likely recovered if multiple edges of the $\mathcal{PK}$ boundary are "pinned" to the same spatial origin in the oracle grid. We define a filtered anchor subset, retaining an anchor α only if its projected top-left origin (t_x, t_y) is geometrically corroborated by an anchor originating from at least one other distinct boundary edge.

In certain $\mathcal{PK}$-derived boundary configurations, this cross-edge validation helps eliminate "phantom" matches—instances where a boundary string matches the oracle by local coincidence but does not align with the global structure of the target. Conversely, in other cases, 2D spatial filtering can degrade attack efficacy by erroneously discarding valid anchors. To rigorously bound the worst-case security, our simulation leverages the Omniscient Threat Model to compute both the filtered and unfiltered candidate masks, dynamically selecting the mask that yields the maximum entropy reduction against the true $\mathcal{SK}$.

AIS Vulnerability Metric Calculation. To compute the exact conditional joint entropy differential $\Delta H = H_{base} - H_{attacker}$ resulting from the application of the candidate bitmask $\mathcal{M}$, we utilize a two-phase Annealed Importance Sampling (AIS) routine:

1. **Base Phase:** We calculate the baseline joint entropy H_{base} of the $\mathcal{PK}$-conditioned glassy phase using standard AIS to establish the underlying cryptographic security.
2. **Constrained Phase:** We execute a modified heat bath MCMC kernel where the tile selection at each spatial index i is strictly restricted to the valid candidates present in the attacker's mask $\mathcal{M}_i$. The resulting constrained joint entropy is defined as $H_{attacker}$.